

Evidence of social learning across symbolic cultural barriers in sperm whales

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This **important** study presents novel data on variation in sperm whale communication, contributing to a richer understanding of the social transmission of vocal styles across neighbouring clans. The evidence is **solid** but could have been further improved with clarification of the specialized metrics and terminology used, particularly for comparisons to other taxa. This research will be of interest for bioacoustics and animal communication specialists, particularly those working on social learning and culture.

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Abstract

We provide quantitative evidence suggesting social learning in sperm whales across sociocultural boundaries, using acoustic data from the Pacific and Atlantic Oceans. Traditionally, sperm whale populations are categorized into clans based on their *vocal repertoire*: the rhythmically patterned click sequences (codas) that they use. Among these codas, *identity* codas function as symbolic markers for each clan, accounting for 35-60% of codas they produce. We introduce a computational method to model whale communication, which encodes rhythmic micro-variations within codas, capturing their *vocal style*. We find that vocal style-clans closely align with repertoire-clans. However, contrary to vocal repertoire, we show that sympatry increases vocal style similarity between clans for *non-identity* codas, i.e. most codas, suggesting social learning across cultural boundaries. More broadly, this subcode structure model offers a framework for comparing communication

systems in other species, with potential implications for deeper understanding of vocal and cultural transmission within animal societies.

Introduction

Cultural transmission is broadly defined as the transmission of information or behaviors between individuals of the same species by means of social learning [1, 2]. While humans represent a benchmark of such capacity, cultural transmission has been observed in a wide variety of animals, including cetaceans [3, 4], songbirds [5], non-human primates [6], and insects [7]. It typically takes one of three forms: vertical transmission, from adult kin to young kin; oblique transmission, from unrelated adults to young; or horizontal transmission, from peer to peer [8].

When animals have the capacity for social learning, group-specific differences can arise and remain stable when they become distinguishable by symbolic markers: arbitrary group-identity signals that are recognizable by both members of the group itself and by members of other groups [9, 10]. In humans, symbolic markers can take a myriad of forms, ranging from visible signs, such as tattoos or garments, to communication cues or signals, such as idiomatic sentences or accents [9, 10, 11]. In animals, however, quantitative evidence of symbolic markers is remarkably scarce, one exception being recent results on the use of *identity codas* in sperm whale social communication [12].

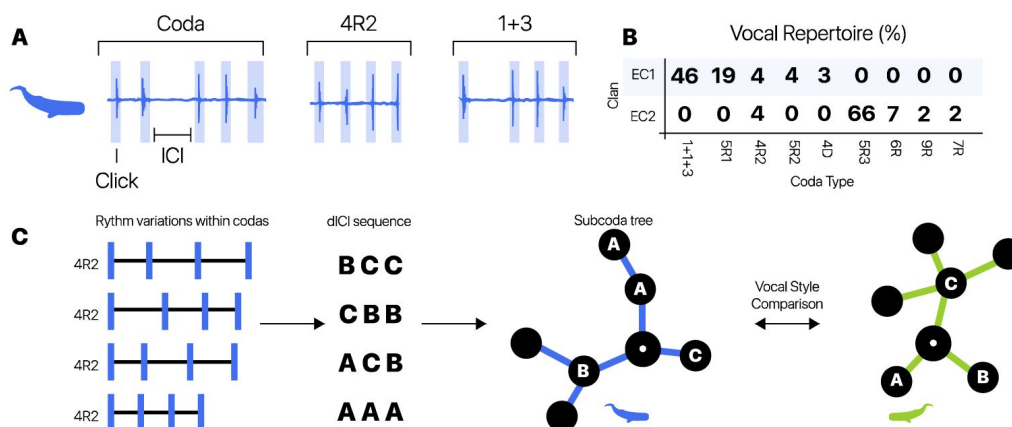


Figure 1

Statistical modeling of subcode structure in sperm whales.

A Sperm whale communication consists of rhythmic sequences of clicks, called codas. A coda is specified by a sequence of inter-click intervals (ICIs). Codas are classified into types based on their rhythmic pattern, which can have various degrees of regularity (e.g., 4R2 vs 1+3). B Social groups of sperm whales employ specific vocal repertoires: the set of coda types they use and their associated usage frequencies. As an illustration, we show those of the EC1 and EC2 clans from the Dominica dataset [13]. Only the most numerous coda types are shown: the rest of the vocal repertoires consists of more coda types with residual frequencies. C The subcode structure can be modeled by considering rhythmic variations within codas of the same coda type. To do so, codas are represented as sequences of discrete inter-click intervals (dICIs), by discretizing absolute ICIs into discrete bins, which can then be considered akin to symbols (e.g. A, B, C, ...), providing a tokenization of codas. Different instances of a single coda type can correspond to slightly different dICI sequences. The resulting dICI sequences are modeled using variable-length Markov chains, which can be represented as *subcode trees*. These trees are built starting from the root node. Each node represents a dICI and a dICI is a child of another in the tree if its presence in the affects the output

probability of the parent. That is the probability of the parent depends on whether the child node has been observed previously in the sequence; see the Methods for a thorough explanation. These trees can be built for an individual speaker or for a group of speakers (here represented by colors), and capture the statistical properties of rhythm variations within codas and in the transitions between those. In other words, the tree captures a vocal style—how they say what they say. The vocal styles of different groups of sperm whales can be quantitatively compared by calculating a distance between their subcoda trees, which is a statistical distance between the dICI output distributions.

Sperm whales live in multi-tiered societies and have a complex vocal communication system [14]. They communicate through rhythmic patterns (*codas*) of short broadband sounds (*clicks*), which have traditionally been classified into a set of *coda types*. This classification is based on the total number of clicks and their rhythm and tempo extrapolated from the time interval between consecutive clicks: the inter-click interval (ICI) [15, 16] (Fig. 1A). For example, the 4-regular (4R2) type refers to a pattern of four evenly spaced clicks, whereas the 1+3 type refers to two clicks separated by a longer pause followed by two clicks in quick succession. Coda types are thus standardized rhythmic patterns, but individual vocalizations of a given coda type exhibit micro-variations around that pattern. In the case of sperm whales ICIs can be used instead of inter-onset intervals (IOIs) due to the very low variance in duration of the individual clicks [17, 18].

The set of vocalized coda types (*coda usage*) combined with how frequently each is vocalized (*coda frequency*) makes up a *vocal repertoire* (Fig. 1B). For example, the 4R2 coda is used by many sperm whales, but other coda types are more specific in their usage or frequency to certain groups of sperm whales. While there is evidence of individual variation in vocal repertoires [19, 20, 21], sperm whales belonging to the same social unit—a stable, matrilineally-based group of whales, share a common vocal repertoire that is stable across years [20, 21, 22]. Social units that share substantial parts of their repertoire are said to be part of the same *vocal clan* [23, 24]. There is clear social segregation between members of different clans, even when living in sympatry, and thus clans mark a higher level of social organization, which appears to be defined on the basis of cultural vocal markers [23, 24, 12] (see Table 1 for a summary of the key concepts).

The clan specific and frequent usage of certain coda types, termed *identity codas* [16, 12], align with the expectations for symbolic markers of group membership [10]. Furthermore, quantitative evidence that sperm whales themselves use identity codas as such markers has recently emerged: the more two clans overlap in geographic space, the more different their identity coda usage is [12]. This is consistent with computational models [10] of the evolution of symbolic marking, which predict that differences between cultural norms will be starkest when inter-group interactions are more common (e.g., in boundary or overlap regions).

All remaining coda types have been referred to as *non-identity* (non-ID) *codas* and constitute a very large fraction of sperm whales' total number of coda utterances. In fact, the total number of emitted non-ID codas accounts for more than 6 out of 10 codas (see SM Section 1.1 for the counts per clan and per coda type). This begets the question: if ID codas are used as clan identity signals, what can be said about the remaining 65% of codas?

Here, we introduce a novel descriptive framework that focuses on the *subcoda structure*, that is, the rhythmic micro-variations of intervals between clicks within codas (Fig. 1C). This framework, formally encoded in what we call a “subcoda tree”, captures *how* codas are uttered: a *vocal style*. We find that variations in this vocal style, even for a single coda type, identify an individual's social unit and clan, effectively fingerprinting vocal repertoires. With this, we add a new dimension with respect to previous approaches based on *which* codas are said—vocal repertoires. Thus, we propose a new concept of *vocal identity* of sperm whales that comprises both vocal style and vocal repertoire.

By applying our modeling framework to acoustic data from the Atlantic and Pacific Oceans, we obtain two main results. First, we partition sperm whale populations into vocal style-defined clans, which we find to recapitulate the previously defined vocal repertoire clans. This confirms our method's validity as it reiterates previous studies' results. Second, and crucially, we find that the vocal style of non-ID codas is more similar for more sympatric clans, i.e. clans whose territory overlaps more spatially. In contrast, we do not find an effect of sympatry on the similarity of vocal styles when studying only ID codas. This suggests that geographic overlap induces vocal styles to become more similar between clans, without jeopardizing each clan's acoustic identity signals. Our results strengthen previous results on the use of ID codas as symbolic markers, while supporting cultural transmission and social learning of vocalizations among whales of *different* clans, as predicted by theoretical models [25 [↗](#)].

Results

Subcoda structure captures variability in sperm whale communication

We model the internal structure of codas, in terms of rhythmic variations at the level of clicks, by using variable length Markov chains (VLMCs). Our analytical pipeline is illustrated in **Fig. 1C** [↗](#). We build each VLMC in two main steps. We first convert codas, naturally represented as sequences of continuous, absolute, inter-click intervals (ICIs), to sequences of discrete ICIs (dICIs), by discretizing time into bins. In this way, each dICI represents a narrow range of possible ICI values. The bins have a fixed width (or resolution) δt and thus implicitly correspond to the temporal resolution of our representation (see Methods for details on the optimal choice of δt). Note that although ICIs have units of time (seconds), dICIs are (unit-less) symbols (e.g. A, B, C, etc.), representing multiples of δt (and so the smaller δt , the more the symbols). For example, the shortest ICIs will be mapped to the symbol A whereas longer ones will be mapped to symbols further down the alphabet. Hence, each coda (a sequence of ICIs) is mapped to a sequence of discrete symbols (a sequence of dICIs). The second part of the pipeline focuses on modeling the internal structure of codas in terms of dICI sequences. Essentially, we want to estimate transition probabilities from a dICI sub-sequence to the next dICI. A standard way would be to describe this using k -order Markov chain models, which model the next state (dICI) probability given a past sequence of k -states (dICIs). However, it is possible that specific dICIs depend on sequences of varying length. Despite we do not have direct evidence of unitary blocks in sperm whale communication, one can imagine this effect similarly to what happens with words (e.g., a word beginning with “re” can continue in more ways than one starting with “zy”). To account for the possibility that states (words, dICIs) might depend on sequences of varying length, while also retaining only the most compressed statistical representation of how codas are structured in terms of dICIs, we employ VLMCs.

Concept	Description
Click	Sharp broadband pulses [26]
Inter-click interval (ICI)	Time interval between two consecutive clicks
Discrete ICI (dICI)	Discretization of an inter-click interval given a bin size.
Coda	Stereotyped, short series of clicks produced by sperm whales in social situations [15]
Coda type	Categorical classification of codas based on rhythm and tempo of coda inter-click intervals [19]
Coda frequency	Usage frequency of each coda type
Coda sample	Set of codas recorded during one event
Coda usage	Set of coda types used by a group of whales
Identity codas	Subset of coda types most frequently used by a sperm whale clan; canonically used to define vocal clans.
Non-identity codas	Coda types produced by whales belonging to multiple vocal clans (i.e., all coda types that are not identity codas)
Vocal clan	Higher-order level of social organization delineated based on social units that share a substantial part of their vocal repertoire [12, 23]
Vocal repertoire	Coda usage and frequency combined. Description of what sperm whales vocalize.
Vocal style	Encodes the rhythmic variations within codas. Description of how sperm whales emit codas
Vocal identity	Concept comprising both vocal style and vocal repertoire

Table 1

Summary of key concepts

VLMCs are generalizations of standard (fixed-memory) Markov chains that allow sub-sequences of dICIs of variable lengths. Longer sequences are kept only if they are significantly more informative in predicting the next dICI than random chance, yielding an optimally compressed representation (see Methods for details on model fitting and selection, including the optimal choice of $\delta > t$). Furthermore, VLMCs naturally have a tree structure (see **Fig. 1C**), because of the natural order between sequences and their sub-sequences. In particular, each node represents a sub-sequence of dICIs, and is equipped with a probability distribution of transitions to the next dICIs. The origin node corresponds to the empty sequence, leaf nodes correspond to the longest sequences, and all nodes forming the branch in between correspond to the sub-sequences of that leaf node. Thus, we call VLMCs fitted to coda ICI data *subcoda trees*.

Note that dICI sequences encode rhythmic variations within codas. Indeed, a coda type is a standard rhythmic pattern that can be realized with variations in its ICIs and thus in its dICIs too. For example, the 4R2 coda type can be vocalized as BCC but also as CBB (in a representation with, say, 26 symbols). In that sense, subcoda trees, through the dICIs sequences that they contain and their transition probabilities, capture information about a vocal style. Four more features of subcoda trees are noteworthy: (i) because the method's input is a set of codas, we can build subcoda trees for repertoires corresponding to different social scales, from individual sperm whales, to social units, all the way up to vocal clans; (ii) the difference between different subcoda trees can be measured using a probabilistic distance (see Methods), which we can use to compare subcoda trees across sperm whale clans; (iii) certain features of the vocal style can be quantified via metrics on the subcoda tree: for example we can define a complexity of the vocal style measured by an entropy on the tree; and (iv) subcoda trees can also be used as generative models, to create new synthetic codas in the form of dICI sequences to train downstream machine learning models.

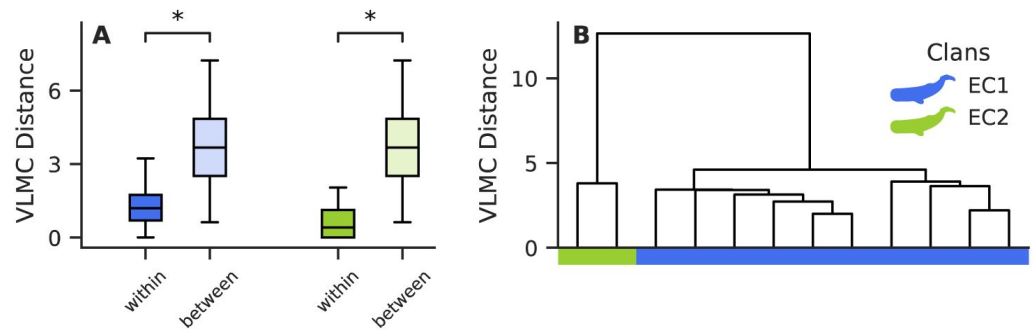


Figure 2

Vocal style recovers social structure of vocal clans in Dominica sperm whales.

A We show the similarity of vocal style, measured as subcode tree-distance, among social units within a vocal clan (*within*, darker color shade) and between two clans (*between*, lighter color shade). We used the manual clan assignments from [24] as ground truth. Vocal style is more similar within clans than between clans. **B** We show the hierarchical clustering of social unit subcode trees. Each leaf corresponds to a social unit, and the colors below show their known clan assignments. The clustering recovers the two-clan structure observed in past work [24].

Vocal style recovers vocal clan structure

The information about vocal style contained in subcode trees is sufficient to recover the social structure of sperm whales (social units and clans). We show this in two ways. First, we analyze a dataset from sperm whales in Dominica (Dominica dataset) [24]. This dataset has rich annotations (code type annotations, identity of recorded whales, social relations of recorded whales) which makes it particularly useful for validation. Specifically, the sperm whales in the Dominica dataset are divided into well known social clans, each composed of several social units, each with its own specific vocal repertoires, and thus can be defined as two different vocal clans. For each social unit in this dataset, we aggregate the individual whales' code samples and build a subcode tree. Computing the distance between these trees (see Methods), we find that the distances between social units within the same clan are significantly smaller than between clans (Fig. 2A). We also find that an agglomerative clustering (average linkage, see Methods for details) on the distance between the subcode trees correctly clusters social units into their respective clans (Fig. 2B). Without a priori knowledge of the clan memberships, we used vocal style to recover the existing classification of social units into two clans, which was previously done based on similarity between vocal repertoires (i.e., code types and usage) [24].

Second, we find that the subcode structure of synthetic codas, generated from subcode trees fitted on real data, closely reproduces that of real codas. To do this, we first train a simple classifier to assign codas to one of the two vocal clans, based on code type. Variations of the same classifier, trained on the same real data, have been shown to discriminate between individual whales, social units, and clans with high accuracy [13]. We train the classifier on real codas, and then test it on both real and synthetic ones. The synthetic codas were generated using the subcode tree of each clan, with a number of codas similar to that of the original dataset for a fair comparison (see Methods for details). We find that synthetic codas are correctly classified into their clans with an accuracy close (~ 85%) to that obtained on the real data (~ 90%, see *Supplementary Materials* Section 4).

Motivated by these results, we extend our analysis to a much larger dataset from the Pacific Ocean (Pacific dataset) [12]. This dataset is more sparsely annotated because of the breadth of its spatial coverage. We restricted our analyses to a well-sampled subset ($n = 57$ coda samples) of the full Pacific dataset (see Methods for details). Coda samples are only labeled by the spatial position at which they were recorded, but no information is available about the identity of the vocalizing sperm whales (see Methods for details). In fact, each repertoire likely contains codas from multiple individuals of a single clan. It has recently been shown that these coda samples can be divided into seven vocal clans based on their coda usage [12]. We use those clans as a benchmark for the following analysis.

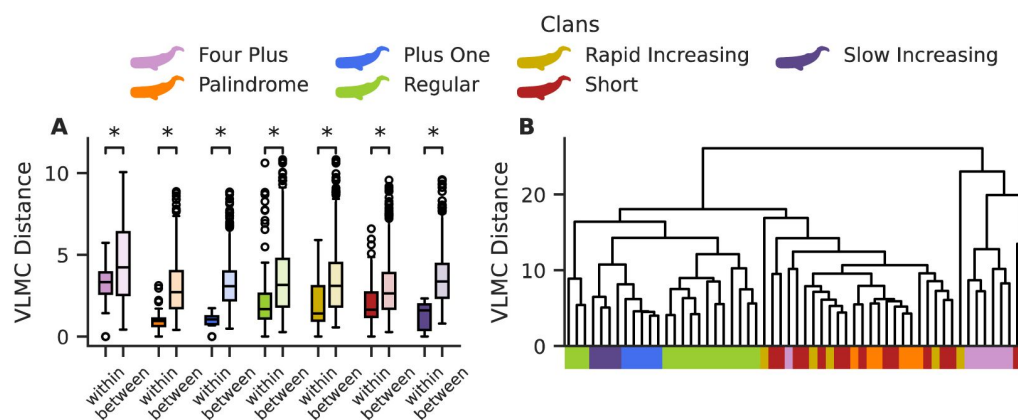


Figure 3

Vocal style recovers social structure of vocal clans in Pacific Ocean sperm whales.

A We show the similarity of vocal style, measured as subcode tree-distance, between coda samples within a vocal clan (*within*, darker color shade) and between a clan and all others (*between*, lighter color shade). We used the vocal clans identified in [12]. Vocal style is more similar within a clan than between clans. **B** We show the hierarchical clustering of subcode trees. Each leaf corresponds to a coda sample, and the colors below show their vocal clan assignments (based on coda usage) from [12]. We find generally good overlap between the groups obtained from clustering vocal style and those from vocal repertoire, with the exception of the *Short* clan (red) that is somewhat mixed with the *Palindrome* (orange) and *Rapid Increasing* (yellow) clans.

Since there is no social unit-level information for this dataset, we fit a subcode tree for each repertoire (i.e., all of the codas recorded on a single day in a single region). Trees are significantly more similar for coda samples belonging to the same vocal clan than for those belonging to different vocal clans (Fig. 3A). We also find that clustering coda samples based on vocal style returns a dendrogram that closely matches the one obtained from coda usage in [12] (Fig. 3B). The major exception we find is the *Short* clan (red), named because member whales produce short codas with very few clicks, for which anomalous results were previously reported as well [12]. In our case, this is due to the Short clan being less well localized in the space of trees, while the other clans have well-defined centroids (see *Supplementary Materials Fig. 9* for a low-dimensional representation subcode tree metric space).

Therefore, we find that sperm whale vocal clans in the Atlantic Ocean (Caribbean Sea) and Pacific Ocean can be identified by a *vocal identity* that encompasses both clan-specific *vocal repertoire* [24, 27, 23, 12] and *vocal style* as defined in this work.

Clan sympatry impacts vocal style of non-ID codas only

While interesting, the fact that both vocal repertoires and vocal styles discriminate between clans might imply that considering both could be redundant for vocal identity. However, we find that this is not the case when we consider the functional role of ID versus non-ID codas.

More precisely, different clans can share significant portions of their total range, overlapping across large swaths of ocean. Such sympatric clans exhibit a decreasing similarity of their ID coda usage with increasing clan overlap [12]. This means that the more two clans overlap in space, the more dissimilar their vocal repertoires are in terms of ID coda types and their usage frequency. This is consistent with the idea that ID codas are used as symbolic markers to delineate cultural boundaries between social groups [12, 10]. In contrast, non-ID coda usage do not show any relationship to clan overlap.

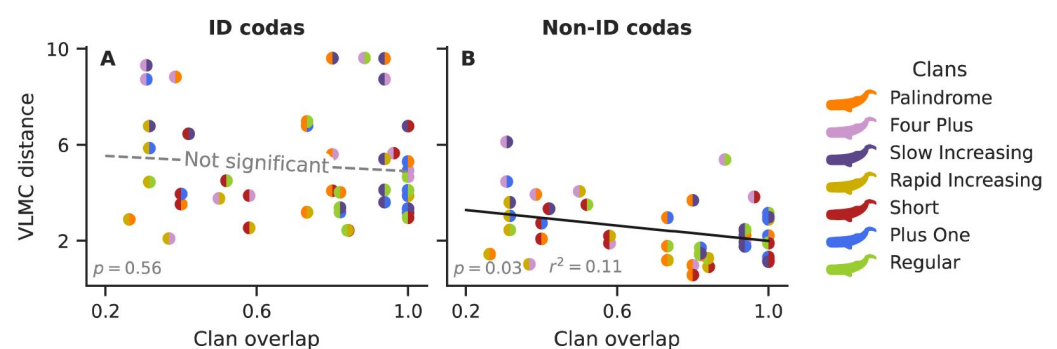


Figure 4

Clan overlap influences non-ID coda vocal styles

Comparing the similarities of different VLMC models fit for each Pacific Ocean clan for both ID and non-ID coda samples. The y-axis represents the measured distance between the subcode trees, and the x-axis shows the geographical clan overlap (as calculated in [12]; see “Measuring clan overlap” in our Methods for details). Each point represents a pairwise comparison between two clans. The effect of overlap on ID coda vocal style similarity is minimal and non-significant while the opposite is true for non-ID codas: overlapping clans produce non-ID codas with a more similar vocal style. The VLMC distances are also typically much greater for ID codas than for non-ID codas. Note that while these results are visually opposite to those reported in Hersh et al. [12], they support the same final conclusions (see “Identity and non-identity codas show different trends” for details).

We find the exact *opposite* effect when considering vocal style. The similarity in vocal style for ID codas across clans does not depend on the level of clan overlap (Fig. 4a). In contrast, the similarity in vocal style for non-ID codas displays a clear and significant increase (i.e., decreasing subcode tree-distance) as clan spatial overlap increases (Fig. 4b). In the *Supplementary Materials* (see Section 2.4.2), we show that the same results hold at the single coda type level, in addition to the whole clan level, along with an analysis of the confidence intervals. These results imply that the internal structure of codas is more similar for groups that likely spend more time in the same space, which can be thought to be analogous to accents aligning in human populations that share the same territory [28, 29]. This also highlights the complementarity of vocal repertoire and style: the trends are different precisely because the two concepts describe different aspects of whale vocalization.

Discussion

We have presented a general method for modeling animal communication systems and their complexity based on VLMCs. In the context of sperm whales, this new method allows the extraction of *subcoda trees*, which succinctly describe the internal temporal structure of codas. Previous work on the structure of sperm whale communication has largely focused on supra-level coda analyses: for example, by classifying codas into types, quantifying how often different types are used, and distinguishing between individual whales, social units, or clans based on those counts [21, 30]. Here, we adopted a more fine-scale approach by investigating potential structure *within* codas. To do so, we used VLMCs to model the transition probability of observing a specific ICI given the previous ones. A VLMC, or here a subcoda tree, encodes all those probabilities but only for dICI sequences that are informative—other sequences are automatically discarded. As such, a subcoda tree is a statistically validated representation of the internal memory structure of codas at the level of sequences of clicks. It contains information about important rhythmic variations and transitions between them: a vocal style.

Using such representations, we propose a novel concept of *vocal identity* for sperm whales composed of *vocal repertoire* (what they say) and *vocal style* (how they say it), the latter being captured by our framework. We find that: (i) vocal styles vary between social units and clans, and can be used to distinguish them; (ii) the similarity of clan vocal styles for non-ID codas increases with increasing spatial overlap, while no change occurs for ID codas; and (iii) social learning across symbolic cultural boundaries most parsimoniously explains the observed trends.

Vocal style recovers hierarchical social structure

Using the Dominica dataset, sperm whales had previously been divided into two vocal clans, based on their vocal repertoires and observed social interactions [24]. In our study, comparing the vocal styles of those same whales led to the same assignment of social units to two vocal clans. Similarly, for the Pacific dataset, clustering based on vocal styles yielded clans that were in good agreement with those previously defined based on vocal repertoires [12] (*Supplementary Materials* for an extended comparison). The difference between the two partitions was mainly due to the Short clan, which was more spread out in subcoda tree space than the other clans, causing overlap with other clans that showed less variability. This variability could be linked to the fact that Short clan whales typically make codas with very few (e.g., three or four) clicks, leading to subcoda trees with very few nodes. In Ref. [12], the authors observe a similar lack of uniformity in coda usage of the Short clan.

Identity and non-identity codas show different trends

For ID codas, we show that the similarity between clan vocal styles is not affected by spatial overlap, while it has recently been shown that the similarity between clan vocal repertoires decreases with overlap (measured on the two clans' joint set of ID codas) [12]. This means that spatial overlap does not affect *how* whales produce ID codas (in terms of their fine-scale rhythmic structure; our results) but does affect *how often* they produce them [12]. In contrast, for non-ID codas, we show that the similarity between vocal styles increases with spatial overlap between two clans, while no change was observed for vocal repertoires in previous work on the same dataset. In other words, increasing spatial overlap is correlated with more similar fine-scale rhythmic structure of non-ID codas produced by whales from different clans (our results), but does not affect how often non-ID codas are produced. Our study thus supports and nuances the results of Hersh et al. [12]. We provide further support for selection acting to produce unambiguous, recognizable identity signals in the ID codas. However, ID codas only account for 35% of the total vocalizations; the remaining 65% of codas have traditionally been lumped into a catch-all category

(i.e., non-ID codas) and their function remains enigmatic (these numbers are an average over the Pacific clans, and go up to 93% for non-ID codas when counting number of coda types instead of number of codas emitted, see SM 1.1 for details). We could still discriminate among clans using non-ID coda vocal style; however, the increased similarity of non-ID coda vocal styles between clans with greater spatial overlap, as demonstrated here, suggests that non-ID codas are likely vocal cues and not identity signals like the ID codas. Accordingly, vocal repertoire and vocal style capture different and complementary information on sperm whale communication, and should be considered in tandem in future studies.

Evidence for social learning across cultural boundaries

There are several potential mechanisms driving the similarity in non-ID coda vocal styles—but not ID coda vocal styles—across spatially overlapped clans: environmental variation, genetics, and/or social learning.

Local adaptation to specific ecological conditions can lead to geographic variation in acoustic signals [31]. If environmental pressures alone were responsible for the trends we observe in sperm whales, this would imply that (i) more spatially overlapped clans experience more similar environments, (ii) non-ID coda vocal style is impacted by or dependent on environmental parameters, and (iii) ID coda vocal style is not impacted by/dependent on environmental parameters. Although the first point is somewhat intuitive, to date there is no evidence that coda production systematically varies with environment. In fact, clans are recognizable across ocean basins, making local adaptation an unlikely driver of the observed trend in non-ID coda vocal style.

If genetic relatedness were responsible, this would imply that (i) more spatially overlapped clans are more genetically related, (ii) non-ID coda vocal styles are genetically inherited, and (iii) ID coda vocal styles are not genetically inherited. If all three requirements were met, then the observed similarity in non-ID coda vocal styles for more spatially overlapped clans could be due to genetic determination under a general isolation by distance structure. However, research to date suggests this scenario is unlikely. Rendell et al. [32] found little evidence to support genetics as an explanation of differences in vocal dialects among clans in the Pacific Ocean. Furthermore, Alexander et al. [33] found that regional genetic differentiation in the Pacific Ocean is very low: while social group is important for explaining both mitochondrial and nuclear DNA variance, geographic region is not. This contrasts with results from the Indian Ocean, where region was the strongest predictor of mitochondrial DNA variance. Given that gene flow in sperm whales is largely male-mediated and that mitochondrial DNA haplotypes are broadly shared across the Pacific Ocean, it is unlikely that coda dialects are genetically determined [33, 34]. Agent-based models grounded in empirical data from Pacific Ocean sperm whales further support coda usage as socially learned, not genetically inherited [25]. To fully rule out a genetic explanation for our results, the analyses in [32] could be replicated for ID coda usage and non-ID coda usage separately. This would shed light on whether certain coda types are genetically inherited vs. socially learned, as has been suggested for some humpback whale (*Megaptera novaeangliae*) vocalizations [35].

The most parsimonious explanation for the observed similarity of non-ID coda vocal styles of clans with increasing spatial overlap is social learning across clan boundaries. This is remarkable, given that sperm whales belonging to different clans have rarely been observed physically interacting at sea [4]. However, that does not preclude the possibility that they are within acoustic range of each other [36] and that cross-cultural social learning opportunities arise. This explanation is compatible with (and bolsters) past work suggesting that ID and non-ID codas function differently in sperm whale communication, and further suggests that they experience different evolutionary pressures [25]. Whether social learning has facilitated stochastic (i.e., cultural drift) or deterministic (i.e., cultural selection) processes is more difficult to determine, and it is unclear whether the observed non-ID coda vocal style alignment has been neutral or adaptive [37],

31 [\[38\]](#)]. Importantly, these findings suggest that vocal learning in sperm whales may not be limited to vertical transmission from related adults to young kin, but that horizontal and/or oblique social learning from outside the natal social unit might also be occurring.

Vocal identity in sperm whales is thus consistent with both cultural selection on ID codas to maintain discrete signals for vocal recognition in sympatry, and social learning between clans leading to a vocal style more similar to that of other whales with which they are in acoustic contact more frequently. This highlights a more complex system of transmission in which clan identity is maintained through selection, while gradual change over time may occur within and across clans for vocalizations which do not function in social recognition and thus may create similar vocal styles. Indeed, the observed alignment of non-ID vocal styles may reflect a combination of automatic imitation, and controlled imitation, guided by social and ecological factors, as discussed in studies of social learning mechanisms [\[38\]](#), [\[39\]](#). In the context of our findings, the similarity in non-identity codas across clans with geographic overlap may indicate an automatic component of social learning. Whales sharing the same habitat are exposed to overlapping acoustic environments, which could facilitate the development of similar vocal styles described here. However, the persistence of distinct identity codas within clans may suggest a more directed form of social learning, shaped by social goals such as clan recognition and cohesion.

Future directions

Our results can be expanded in multiple ways in future work. The first, and the simplest conceptually, would be to conduct the present analysis on a larger dataset. More codas would improve the quality of the statistical analyses and ensure that all codas are represented in realistic proportions for each clan. Moreover, longitudinal datasets might provide direct evidence to discriminate between the social learning hypothesis and competing ones (e.g. drift in vocal style). Similarly, confirmations could emerge from large scale genetic datasets addressing the issues of phylogenetic relatedness (or lack thereof) in clans that are closer in vocal style distance. Such datasets do not exist at present, but efforts towards automated and semi-automated collection techniques are underway (e.g. Project CETI [\[40\]](#)). Second, from a methodological perspective, we could add spectral information (in terms of acoustic frequencies) to the temporal information currently used. Although sperm whale acoustic communication seems mostly based on rhythm, spectral features of individual clicks may convey additional information. This possibility could be incorporated into our method by labeling the dICIs according to the frequency content of the associated click (or by extending the available “alphabet” for the VLMC). Third, it would be interesting to investigate in more detail the function of non-ID codas. Indeed, even though ID codas were only recently formally named for the first time, they have been the primary focus of sperm whale coda research for decades. As previously mentioned, non-ID codas are a catch-all category for anything that is not an ID coda, but that does not mean that all non-ID codas function in the same way. To start to unveil their function, we need to consider the context (behavioral, environmental, etc.) in which different non-ID codas are produced [\[41\]](#). The pattern we documented may or may not apply to all non-ID codas, but it is at least strong enough that we detect the relationship with clan spatial overlap when collectively considering all non-ID codas. Finally, the methods we developed here could in principle be used to capture the structural complexity of any communication system, including those coming from other animal species, provided they can be encoded in sequences of tokens (e.g. calls, signs, etc.). This would also allow to compare the complexity of communication systems across species, possibly linking it to other ecological parameters of the environments inhabited by such species.

Methods

Acoustic data

We analyzed two datasets in the present study. The Dominica dataset contains 8719 annotated codas recorded in the Atlantic Ocean off the island of Dominica between 2005 and 2019. The codas come from 12 social units grouped into two vocal clans (EC1 and EC2). The Pacific dataset was collected between 1978 and 2017 at 23 locations in the Pacific Ocean (the recording methods are available in the supplementary materials of [12]). The codas were divided into coda samples according to their recording day and each repertoire was assigned a single vocal clan inferred in [12]. When considering a clan-level analysis (Fig. 3) all coda samples were used to compute the subcoda trees (23555 codas). However, when analysing at a coda samples level (Fig. 4), we discarded coda samples with less than 200 codas with statistical inference in mind, resulting in a final count of 57 coda samples (17046 codas) for the Pacific.

Representation of sperm whale communication as discrete inter-click intervals

As a preliminary step, we discretized the (continuous) ICI values into bins of width δt seconds. In other words, we represented the continuum of ICI values by a finite set of discrete ICIs (dICIs) based on the duration of the ICI. The bin width δt controls the *temporal resolution* of the representation: a higher value of δt implies a coarser representation with fewer dICIs. We also imposed an upper bound $t_{\max}$: any ICI value greater than that was truncated to $t_{\max}$. This ensured that the set of dICIs was finite. Note that although ICIs have units of time (seconds), dICIs are unitless (they represent time intervals). The resulting representation of ICIs as dICIs is a discrete random variable defined as

$$X_{\delta t, t_{\max}} = \left\lfloor \frac{\min(\text{ICI}, t_{\max})}{\delta t} \right\rfloor, \quad (1)$$

which takes values in the finite set $\chi = \{0, 1, \dots, \lfloor \frac{t_{\max}}{\delta t} \rfloor\}$. We represented the sequences of ICIs by sequences of dICIs from that finite set. Note that any ICI value above $t_{\max}$ is mapped to the dICI $\lfloor \frac{t_{\max}}{\delta t} \rfloor$ and therefore represents the end of a coda. We set $t_{\max} = 1$ (longer than any ICI) and $\delta t = 0.05$ throughout the analysis (see *Supplementary Materials* section 3.3.2 for justification of this choice and section 3.4.3 for an analysis on the influence of this parameter).

Variable length Markov chains

We then modeled these dICI sequences using variable length Markov chains (VLMCs). VLMCs provide the large memory advantage of higher-order Markov chains when needed, without the drawback of having too many unnecessary parameters in the model.

Fitting a VLMC is the process of deciding how much memory is necessary to model specific sequences. The criterion for making this decision is the following: longer sequences are discarded if their distribution of transition probabilities is similar to that of shorter subsequences. This process is often called *context tree estimation* and consists of two steps.

The first step is to consider W_D the set of all sequences of maximum length D (which we set to 10) and to assign the following probability distribution q_w to each sequence:

$$q_w = P(X|w), \quad (2)$$

that is, the probability of observing a state $x \in \chi$ given the sequence w .

The second step is to prune the sequences that do not add information. Take two sequences $u, w \in W_D$, one being the suffix of the other $w = \sigma u$. The information gained H_w by considering the longer sequence can be measured with a weighted Kullback-Leibler (KL) divergence D_{KL} [42]. The longer memory sequence w is kept only if the information gain is greater than some threshold K [43, 44].

$$\Delta H_w = N(w)D_{KL}(q_w||q_u) > K \quad (3)$$

where $N(w)$ denotes the length of sequence w . Sequences that satisfy this condition are called *contexts* and sequences that do not are discarded. A VLMC can be defined as the set of these contexts w and their associated probability distribution q_w (see *Supplementary Materials* section 3.1 for details).

A VLMC can be visualized as a tree by representing each context w by a node and setting the root node as the context of length zero. Contexts that are subsequences of each other are then part of the same branches, which end with the longest contexts.

Quantitative Comparison of VLMCs

If two VLMC models T_1 and T_2 are built over the same finite set of dICIs χ , there exists a map $\phi_1 : W_D \rightarrow T_1$ that maps any sequence of elements of χ into the longest sequence present in T_1 , and similarly for T_2 . This map also induces a map between the probability distributions of T_1 and T_2 . Given two distributions over the same set χ , we can measure how different they are with the KL divergence. Therefore, it is possible to define a dissimilarity between T_1 and T_2 by considering the average KL divergence over all sequences of T_1 and their map $\phi_1(T_2) \subseteq T_1$

$$d_{KL}(T_1, T_2) = \frac{1}{|T_1|} \sum_{w \in T_1} D_{KL}(q_w || p_{\phi_1(w)}) \quad (4)$$

Refer to the *Supplementary Materials* section 3.4 for a more detailed explanation.

This results in a dissimilarity measure that captures not just the difference in emission distribution but also the structural differences of the associated context trees. When comparing the distribution of distances in **Fig. 2A** and **Fig. 3A** we performed a *Kolmogorov-Smirnov* test to test if the distances between social units/coda samples of the same clan and distances between social units/coda samples of different clans had come from the same distribution. For every pair, we can reject the hypothesis of the distances coming from the same distribution with 95% confidence.

Hierarchical Clustering of VLMCs

The dendrograms in **Fig. 2B** and **Fig. 3B** were obtained by hierarchical clustering using average linkage on the set of subcoda trees (VLMCs). Since the distance is not symmetric, for agglomerative clustering we considered the symmetric distance:

$$d_{KL}(T_1, T_2) = \max \{d_{KL}(T_1, T_2), d_{KL}(T_2, T_1)\}. \quad (5)$$

Measuring clan overlap

We used the clan spatial overlap values from [12]. Briefly, given two clans A and B, and the coda samples associated to them, the amount of geographical overlap of A in B was measured as the fraction of coda samples belonging to clan A that were recorded within 1000 kilometers of at least one repertoire of clan B. One thousand kilometers is the approximate annual home range span of sperm whales in the eastern tropical Pacific [45, 46].

Statistical Testing

On [Fig. 2](#) and [Fig. 3](#) we compare the distributions of distances between subcoda trees of coda samples/social units of the same clan (*within*) and of different clans (*between*). The purpose is to assess whether these distributions originate from the same underlying population. We employ both the *Kolmogorov-Smirnov* test and the T-test. The observed *p*-values were well below 0.01 for all clans. This allows us to confidently reject the hypothesis that there is no difference between the vocal style between different clans. For more information check the *Supplementary Materials* in section 3.4.2.

To assess the existence of a relationship between clan overlap and vocal style similarity, we applied an ordinary least squares linear regression model (OLS). We show the resulting *p* values of the OLS statistical test at the bottom left of each plot of [Fig. 4](#) along with the observed r^2 value. To assess whether there is true difference between the two cases, we also bootstrapped the linear regression calculation to obtain 95% confidence intervals for the slopes of the fits, resulting in both negative and positive values in the ID case, but only negative slope values for the non-ID case, thus confirming our interpretation.

Supplementary Materials

1. Data and Preprocessing

Sperm whales communicate vocally via *clicks* : short bursts of sound emitted in sequence. These clicks are combined into recognizable patterns called *codas*. Clicking sperm whales were recorded with a hydrophone, and clicks were detected in the resulting audio files by human experts. The data that we used consists of time sequences of inter-click intervals (ICIs), i.e. the times between two consecutive detected clicks—this is equivalent to having a time series of click onsets.

We used two datasets: the Dominica and Pacific datasets. The Dominica dataset contains 8719 annotated codas recorded in the Atlantic Ocean near Dominica. The codas come from 12 social units grouped into two vocal clans (EC1 and EC2). The Pacific dataset consists of around 23555 codas recorded between 1978 and 2017 in 23 Pacific Ocean locations. The codas were divided into coda samples according to their recording day and each coda sample was assigned a single vocal clan inferred in [\[1\]](#). When considering a clan-level analysis all coda samples were used to compute the VLMC models. However when analysing at a coda sample level, we discarded coda samples with less than 200 codas with statistical inference in mind, resulting in a final count of 57 coda samples (17046 codas) for the Pacific.

To model the ICI sequences in the datasets, we represented the ICIs by a finite set of symbols (or states), in three main steps ([Fig. S1](#)). First, we denoted X a continuous random variable that represents an ICI:

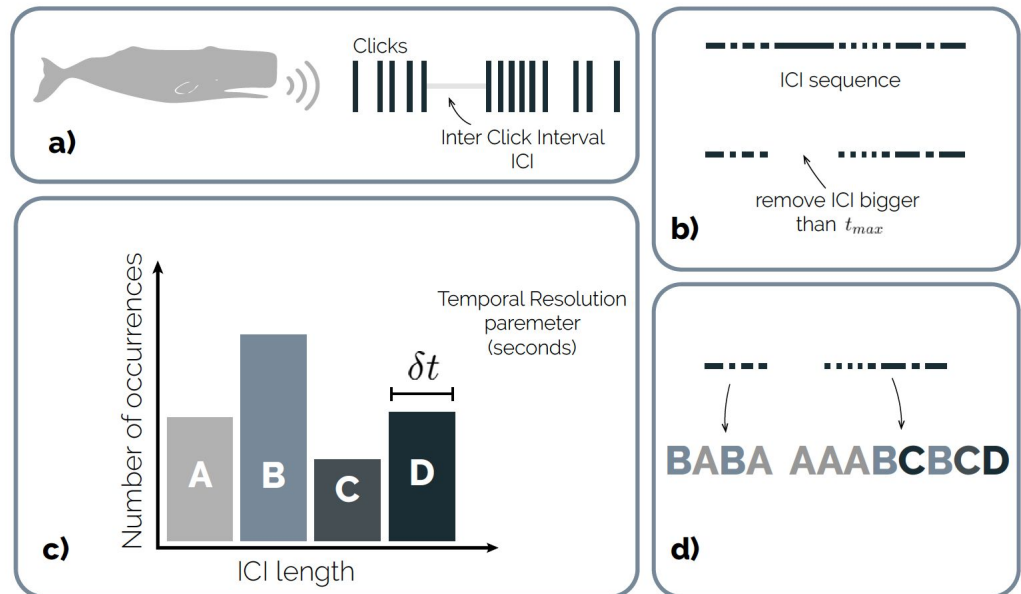
$$X = \text{ICI}. \quad (\text{S1})$$

Second, we imposed an upper bound $t_{\max}$ on the values taken by X . This was to make sure that we are modeling with a finite number of states. Specially, in situations here the maximum number of states have to be known. We set this value to 1 second to be sure that it is longer than any ICI.

$$X_{t_{\max}} = \min(\text{ICI}, t_{\max}). \quad (\text{S2})$$

Thirdly and finally, we discretized the values of the ICIs into a set of bins, akin to a histogram. We denote δt the width of these bins in seconds and call it the *temporal resolution* of the representation. Formally, we define

$$X_{\delta t, t_{\max}} = \left\lfloor \frac{\min(\text{ICI}, t_{\max})}{\delta t} \right\rfloor. \quad (\text{S3})$$



Supplementary Figure S1

Representing sperm whale communication as a sequence of symbols.

a The temporal patterns of click onsets are equivalently encoded as sequences of inter-click intervals (ICIs). **b** We discard ICI values larger than a threshold $t_{\max}$ seconds, and **(c)** discretize the others into a finite set of bins of width δt seconds. Each bin is then assigned a symbol (here a letter of the Latin alphabet), so that each symbol represents an ICI in a given range. **d** We then represent sperm whale communication as a sequence of these symbols.

By construction, this defines a discrete random variable that takes values in the finite set $\chi = \{0, 1, \dots, \lfloor \frac{t_{\max}}{\delta t} \rfloor\}$. Note that each element of this set represents a range of ICI values of length δt seconds. Any ICI value above $t_{\max}$ is mapped to the symbol $\lfloor \frac{t_{\max}}{\delta t} \rfloor$. It thus represents the end of a coda. We set the upper bound to $t_{\max} = 1$ seconds and $\delta t = 0.05$ (see Section 3.3 for details about the choice of values).

We then modeled sperm whale communication sequences $(X_i)_{i \in \mathbb{N}}$. In addition, for clarity, we will denote the elements of χ by letters of the Latin alphabet A, B , and so on. In terms of terminology, we will also refer to χ as an alphabet, and to its elements as symbols or states, interchangeably.

1.1 ID and non-ID codas

Some coda types are considered ID for some clans but non-ID for others. The Pacific dataset [1] has annotations specifying this label (ID or non-ID) for each coda and each clan. For each clan, we counted the total number of codas that are ID and the total number of codas that are not (see **Table S1**).

We also provide similar statistics for the count of different coda types (see **Table S2**).

Clan Name	Total	ID	Non-ID	Percentage of ID
Four Plus (FP)	2590	587	2003	22.66%
Palindrome (PALI)	2124	1249	875	58.8%
Plus One (PO)	2083	1252	831	60.11%
Regular (REG)	8289	3545	4744	42.77%
Rapid Increasing (RI)	1776	425	1351	23.93%
Short (SH)	5428	562	4866	10.35%
Slow Increasing (SI)	1265	541	724	42.77%
Total	23555	8161	15394	34.65%

Supplementary Table S1

Count of Codas

Clan Name	Total	ID	Non-ID	Percentage of ID
Four Plus (FP)	72	2	70	2.78%
Palindrome (PALI)	64	9	55	14.06%
Plus One (PO)	54	6	48	11.11%
Regular (REG)	74	9	65	12.16%
Rapid Increasing (RI)	79	2	77	2.53%
Short (SH)	85	1	84	1.18%
Slow Increasing (SI)	59	3	56	5.08%
Total	487	32	455	6.57%

Supplementary Table S2

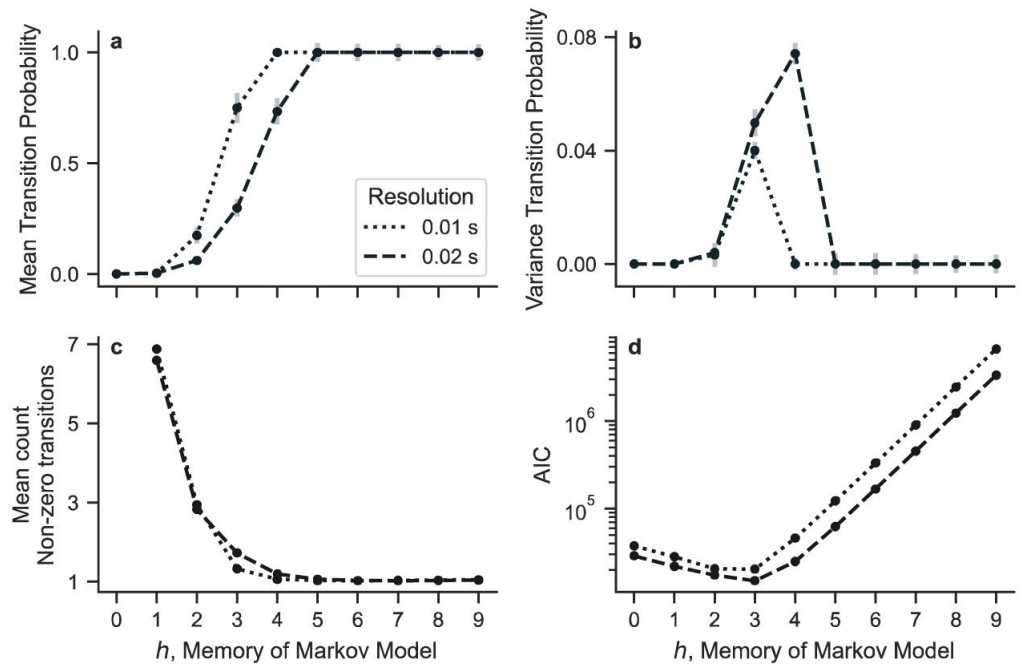
Count of Coda Types

2 The role of memory

A coda is a series of clicks emitted in fast succession. They have traditionally been identified by practitioners as building blocks for sperm whale communication. Whether or not codas themselves are composed by series of smaller collections of clicks is however an open question. We asked: are the codas the smallest such blocks, or is there structure at a scale shorter than codas—but longer than the individual clicks that constitute them? In order to answer this, we modeled sperm whale communication as higher-order Markov chains, that is, Markov chains with a memory h larger than or equal to one (but of fixed length)—see Section 2 for details. In other words, we assume that the probability of observing an ICI within given range—here referred to as symbol or state—depends on the h previous states. For a given h , we fit this Markov model to the data by estimating the transition probabilities from sequence of h state to any other state.

The results are summarized in **Fig. S2**, for a range of memory values h , and for two temporal resolutions for the binning of the ICIs. We note that there is a bifurcation between two different behaviors around $h \approx 3$: transition probabilities go from very low (approximately random transitions) to very large (almost deterministic transitions). Indeed, for $h < 3$, these probabilities are very low [**Fig. S2(a)**] and all similar [**Fig. S2(b)**]. All possible next states are equally likely, but not very likely: this indicates underfitting. On the contrary, for $h \approx 3$, the transition probabilities are all close to one and all similar. Moreover, only a few of them are non-zero [**Fig. S2(c)**]. Given a sequence longer than three, only one next state can be observed: this indicates overfitting. Finally, for $h \approx 3$, transition probabilities are heterogeneous: their average is between 0 and 1, their variance exhibits a peak, and more than one state can potentially be observed next. Moreover, the Akaike Information Criterion (AIC) displays a minimum around that value of the memory, which indicates that it provides a good trade-off between variance of the data explained and the number of parameters needed for the model.

This transition around $h \approx 3$ suggests that there is structure at that level of memory, which is shorter than most coda types. This motivates our search for structure within codas. However, fixed-memory Markov Chains do not allow for different configurations to have different levels of memory, which leads to variable length Markov Chains (described in the next section).



Supplementary Figure S2

Fixed length Markov model bifurcates as a function of memory length h .

We show (a) the mean, and (b) the variance of the transition probabilities as a function of h . We also show (c) the number of non-zero transition probabilities per state, and (d) the AIC. The dashed and dotted lines represent two different temporal resolutions (i.e., bin size used during Preprocessing). There is a bifurcation around $h \approx 3$ that suggests subcode structure.

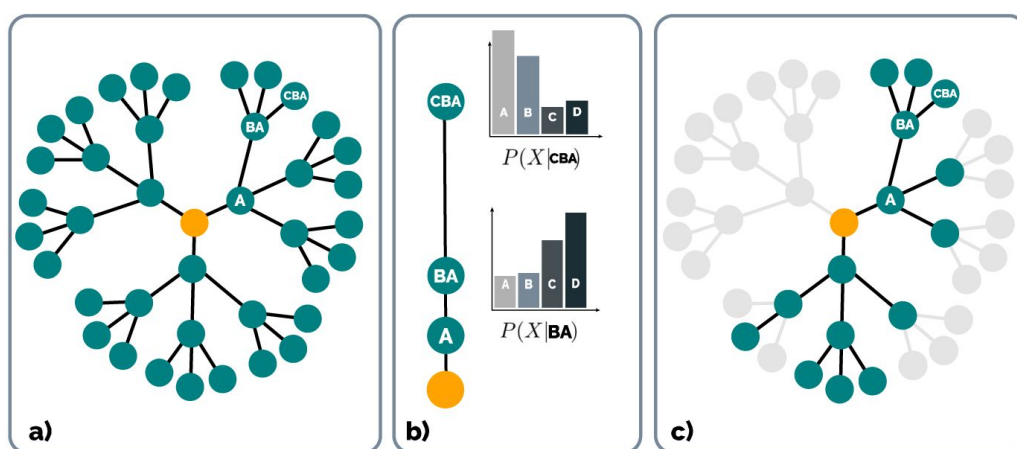
3 Variable Length Markov Chains

Some states can be predicted with more or less memory of past states than others. This observation is the base motivation for introducing variable length Markov Chains (VLMCs) which go beyond the fixed-memory limit of traditional Markov chains. Take for example a state X_2 that has the same probability of occurring knowing the last two states ($x_0x_1 \in \chi^2$) or only the last state ($x_1 \in \chi^1$):

$$P(X_2|x_0, x_1) = P(X_2|x_1). \quad (S4)$$

In this case, a shorter memory ($h = 1$) is sufficient and we do not need a longer one (h_2).

In practice, VLMCs bypass the necessity of having $(n - 1)n^h$ parameters by allowing states to have unequal lengths (memory). Smaller lengths are preferred whenever the additional memory does not significantly change the distribution of transitions to the next possible states.



Supplementary Figure S3

Steps for fitting a VLMC from a sequence.

Step 1 (a): Construction of the full (or saturated) suffix tree up to maximum depth D . Step 2 (b): Assigning a probability measure to every element of the tree. Step 3 (c) Pruning of nodes that carry the same information content as parents according to Eq. (S10). In this example, context CBA is relevant because it changes the distribution of transition probabilities with respect to its parent BA and therefore it is not pruned. The orange node represents the root node.

3.1 Building a VLMC

For a Markov model of fixed order h , the set of possible states χ^h is composed of all possible sequences of length h . For a VLMC, however, states can be sequences of arbitrary length. The set of possible states is thus a subset of the set of all sequences that can be built from the alphabet χ , including the empty sequence $\chi^0 = \emptyset$. Let W denote this set.

$$W = \bigcup_{i=0}^{\infty} \chi^i \quad (\text{S5})$$

In this project, because codas are typically constructed from a small number of clicks, we only consider *finite* length sequences (see [2] for non-finite VLMCs). In practice, we choose a *maximum* memory allowed D , which we set to $D = 10$, much larger than the typical coda length.

$$W_D = \bigcup_{i=0}^{D=10} \chi^i \quad (\text{S6})$$

Fitting a VLMC is the process of finding some subset $L \subseteq W_D$ where the elements satisfy the condition: shorter states are preferred if their distribution of transition probabilities is similar to their longer length equivalents. This is generally called context tree estimation in the literature [3].

Probabilising the tree We start with W_D for some D which we take to be equal to 10. To each element of $w \in W_D$ we assign a probability distribution q_w over the set χ as the probability of observing a state $x \in \chi$ given a previous sequence w .

$$q_w = P(X|w) \quad (\text{S7})$$

Where P denotes the likelihood estimation computed as

$$q_w(X = x) = P(X = x|w) = \frac{N(wx)}{\sum_{c \in \chi} N(wc)} \quad (\text{S8})$$

where $N(w)$ the number of occurrences of the sequence w .

Pruning the tree Given two sequences $u, w \in W_D$, we say that u is a suffix of w if $w = \sigma u$ for some other sequence σ of length ≥ 1 . If $\sigma \in \chi$ we say that u is a parent of w . That is, u is a parent of w if u is a suffix of w and w is longer by only one letter.

In an intuitive way, u is a parent of w if w “looks into the past” one step further than u . At the core of the VLMC is measuring the information gain in using the longer memory u instead of its shorter memory parent w . If this information gain is not sufficient, then we discard the longer memory u . We measure the information gain in using the longer memory w instead of u with a weighted Kullback-Leibler (KL) divergence DKL [4]:

$$\begin{aligned} \Delta H_w &= N(w) D_{KL}(q_w || q_u) \\ &= N(w) \sum_{x \in \chi} P(x|w) \log \frac{P(x|w)}{P(x|u)} \end{aligned} \quad (\text{S9})$$

The longer memory sequence w is kept if and only if the information gain is greater than some threshold K [5, 6]. Refer to Section 3.3 for a discussion on the value of K . The set of all sequences that respect the above threshold are called *contexts*, and denoted by T :

$$T = \{w \in \mathcal{W}_D | \Delta H_w > K\} \quad (\text{S10})$$

A VLMC is the the Markov model with the set of states:

$$\mathcal{L} = \{wx \mid t \in T, x \in \mathcal{X}\} \quad (\text{S11})$$

and with transition probabilities defined by q_w for $w \in T$.

3.2 Model Selection

When modeling a process $(X_i)_{i \in \mathbb{N}}$ with a Markov model, the memory length h controls the trade-off between complexity and error. Higher memory values tend to result in models that generalize poorly. On the other hand, lower values of h fail to capture the patterns, resulting in a uniformly random model.

To choose an appropriate value of h , it is common to employ some statistic that measures the trade-off between precision in prediction and the number of parameters. A model with high predictability and a low number of parameters is favored. There is a wide range of metrics [7] and one of the most widely used is the AIC [8, 9]:

$$\text{AIC} = 2k - 2 \log \hat{P}((X_i)_{i \in \mathbb{N}} | \mathcal{M}_h). \quad (\text{S12})$$

$\hat{P}((X_i)_{i \in \mathbb{N}} | \mathcal{M}_h)$ is the maximum likelihood of the sequence $(X_i)_{i \in \mathbb{N}}$ given the Markov model $\mathcal{M}_h$ with memory length h and k parameters (transition probabilities). The best model is indicated by the lowest AIC.

3.3 Parameter sensitivity

3.3.1 Information gain threshold K

The threshold value K represents the minimum information gain necessary to increment the memory of a given context by one. This value ultimately influences the depth and shape of the VLMC model. Low thresholds result in deep trees with many parameters and are prone to overfitting, whereas small threshold values cause trees to not expand past low values of memory and potentially fail to capture statistical dependencies (Fig. S4).

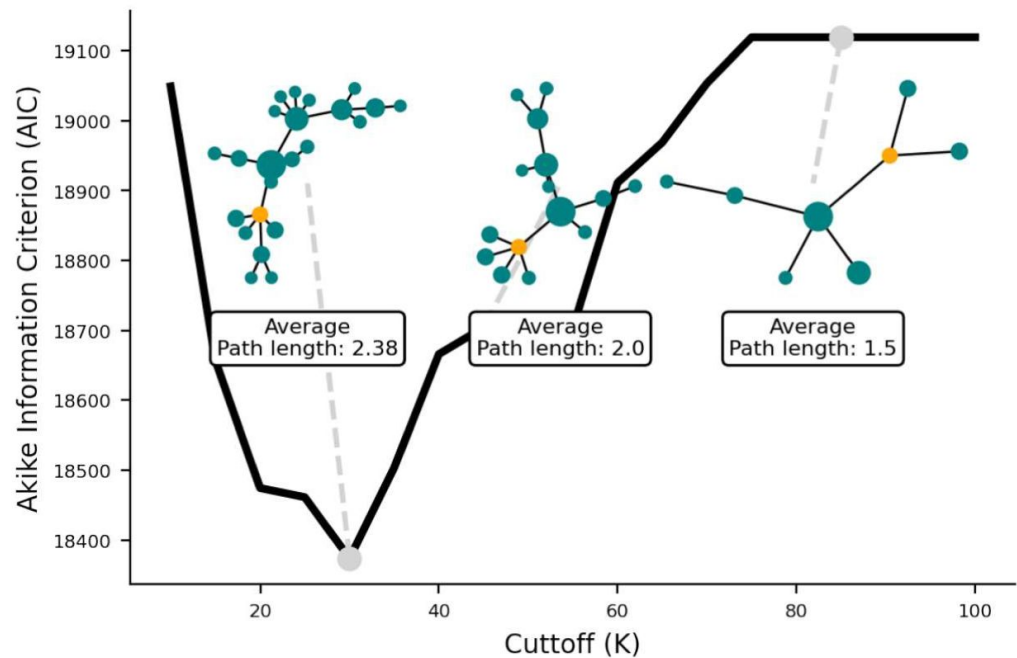
The best value of K should be the one that outputs the optimal VLMC model (i.e., the one that minimizes the AIC).

$$\arg \min_{T \subseteq \mathcal{W}_D} \text{AIC}(T) \quad (\text{S13})$$

However, this search is done over the entire set of possible context trees. The problem of estimating the optimal context tree is an ongoing research area although many good methods have been proposed [6].

For some threshold K . The above dissimilarity is an expression of differences of deviances and as such follows an asymptotic $\frac{1}{2}\chi^2$ distribution [5] with $|\chi| - 1$ degrees of freedom. As such we can set the K thresholds to represent quantiles of a χ^2 distribution. We use the 0.95 quantile, meaning we keep the child whose additional memory exceeds the value:

$$K = \frac{1}{2}\chi^2_{|\chi|-1,0.95} \quad (\text{S14})$$



Supplementary Figure S4

Selecting the optimal information cutoff K .

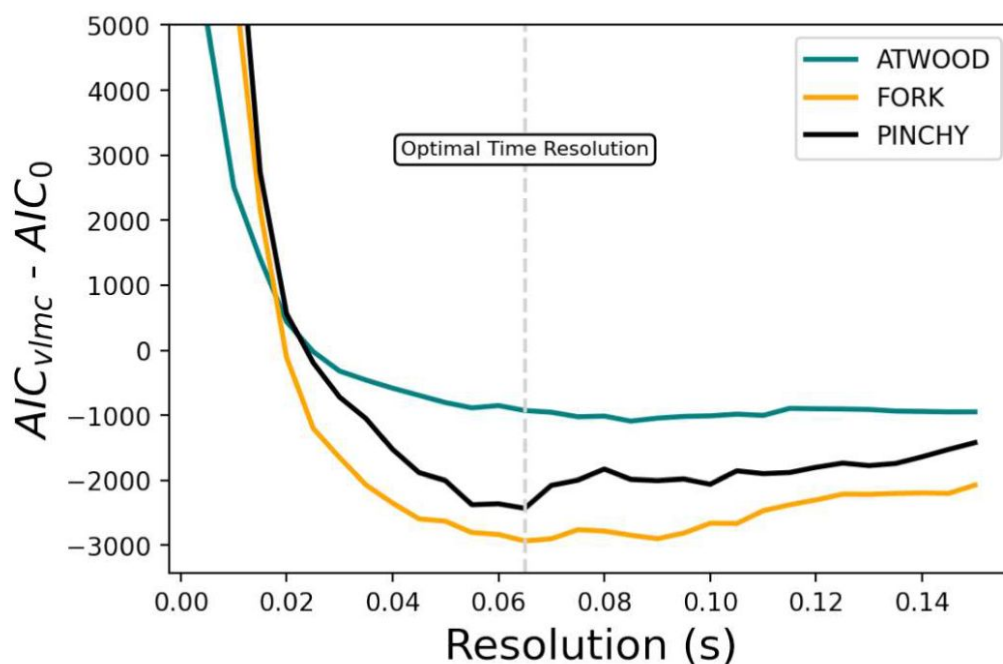
The AIC for different VLMC models with respect to the cutoff value (K) used to build them. Above are some context trees associated with three different VLMCs. The trees decrease in size for the threshold increases since the information gain has to be greater for a context to be “accepted” in the tree. This effect is also visible in the average path length which can be seen as the average memory of the model.

3.3.2 Temporal resolution δt

The temporal resolution δt denotes the scale at which we discretize the continuous, absolute ICI values into bins. A small value might provide differentiation between clicks, but also burdens the VLMC models by increasing the number of parameters and states.

To select the most appropriate resolution parameter, one might be tempted to compare the AIC obtained from different VLMC models extracted from data at different resolutions. However, that is not possible since models fit on different data are not really comparable. Imagine the extreme case with a time resolution so large that all clicks are mapped to the same discrete symbol: any model fit on this data would achieve an optimal AIC value.

In our case, we compare the AIC obtained from the VLMC model with the AIC obtained from a fixed length Markov model of order 0 fitted to the same aggregated data. The intuition behind this approach is that the zero length Markov model represents how “easy” it is to predict the data. The best resolution would be the one where the difference between the AIC of our fit VLMC is the biggest when compared to the 0th order Markov model (Fig. S5 [↗](#)).



Supplementary Figure S5

Selecting the optimal temporal resolution.

The temporal resolution parameter δt for the preprocessing steps is chosen by subtracting the AIC of the 0th order memory Markov model from the AIC of a VLMC model, which encodes how well the model predict versus how easy it is to predict the data. We use this setup because the AIC alone is not sufficient, because the actual fitted dataset changes with the resolution parameter, and thus models with different δt are not directly comparable. The minimum value is the best resolution parameter although any value between 0.01 and 0.1 seems acceptable. We show this for three individual sperm whales (colors) from the Dominica dataset.

3.4 Quantitative comparison of VLMCs

The KL divergence is one of the most used methods for measuring statistical dissimilarity between two distributions, mostly due to its connections to information-theory. Being a generative model, a VLMC is not a single probability distribution, but a set of distributions q_w one for each context $w \in T$.

Given two VLMC models T_1 and T_2 over the same alphabet χ . Let p and q be their associated sets of transitions distributions, respectively. We define the divergence $d_{KL}(T_1, T_2)$ between them as the average KL divergence between the set of associated transition distributions.

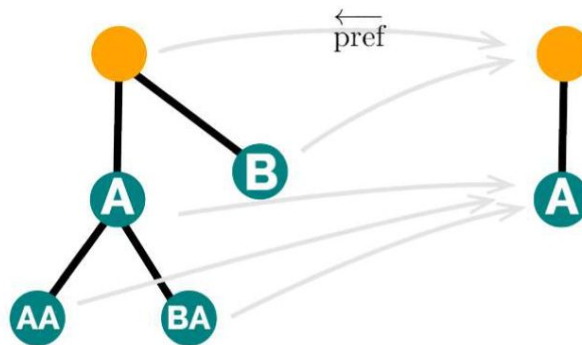
However, there may not be a one-to-one map between q_w and p_w . In fact, more often than not T_1 and T_2 have a different number of contexts. As such, for every $w \in T_1$ we associate $u \in T_2$ where u is the longest suffix of w that belongs to T_2 . This results in a dissimilarity measure that captures not just the difference in emission distribution but also the structural differences of the associated context trees:

Divergence Given two VLMCs $(T_1, (q_w)_{w \in T_1})$ and $(T_2, (p_w)_{w \in T_2})$ built over the finite alphabet χ we define the distance:

$$\begin{aligned} d_{KL}(T_1, T_2) &= \frac{1}{|T_1|} \sum_{w \in T_1} D_{KL} \left(q_w || p_{\overleftarrow{\text{pref}}(w)} \right) \\ &= \frac{1}{|T_1|} \sum_{w \in T_1} \sum_{x \in \chi} q_w \log \frac{q_w}{p_{\overleftarrow{\text{pref}}(w)}} \end{aligned} \quad (\text{S15})$$

Where $\overleftarrow{\text{pref}}(w)$ denotes the longest suffix of $w \in T_1$ that belongs to T_2 .

$$\overleftarrow{\text{pref}} : T_1 \rightarrow T_2 \quad (\text{S16})$$



Supplementary Figure S6

Measuring dissimilarity between two VLMC models.

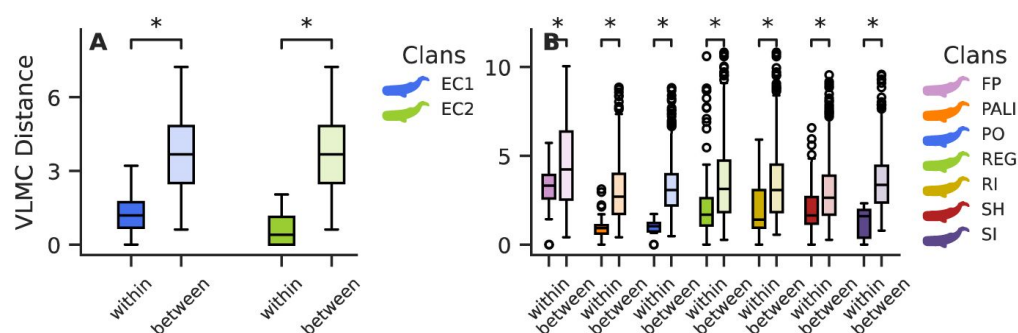
The two trees represent two VLMCs models built over the same alphabet χ . We measure their dissimilarity or distance with the KL divergence.

3.4.1 Statistical testing on distribution of the distances

We fit a VLMC model on coda samples/social units from different clans on both the Pacific and Dominica dataset. For each clan we compute the distances between all VLMC models belonging to that clan (*within*) and between the models of the clan to the ones belonging to other clans (*between*) (Fig. S7). On each pair (*within/between*) we tested the distributions to check if they both came from the same populations. We employed both the *Kolmogorov-Smirnov* test and the *T*-Test. We also measured the effect size using *Cohen's* method. The resulting statistics for both datasets can be found in the tables below.

3.4.2 Non-ID results by coda type

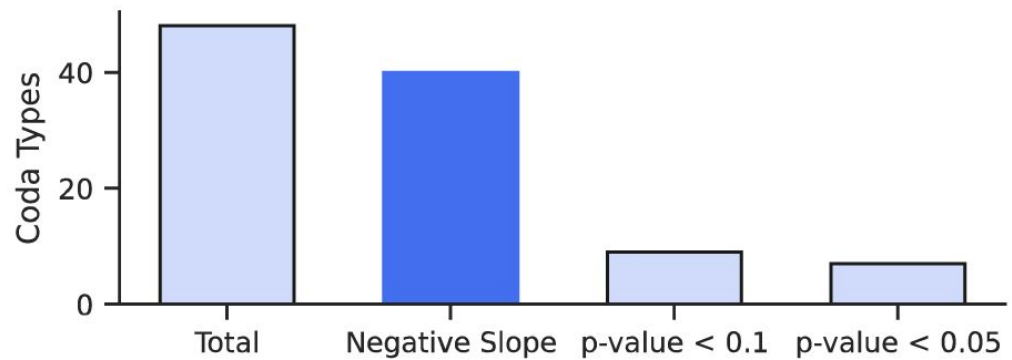
In this section we repeat the approach on comparing the geographical clan overlap with the VLMC distance on non-ID codas. In contrast to the main text, we segment each set of codas by coda type and note the slope, the p-value of the Pearson correlation and the p-value according to a Spearman correlation, Table S4. Just as the main text, when segmenting by coda type we observe that the vast majority of correlations is negative, i.e., geographically overlapped clans have a more similar communication. However, although most correlations are negative, only a small portion is significant. It is important to take into consideration that the amount of data used to fit each VLMC model is considerably reduced given the extra segmentation. Furthermore, coda types that were uttered exclusively by only two clans were also omitted as it is always possible to draw a line between two points, and thus a linear analysis makes little sense.



Supplementary Figure S7

Comparison of the distribution of distances between VLMC models, with statistical significance.

Each color represents a clan of a given dataset: **A)** Pacific, and **B)** Dominica. The *within* distribution represents the distance between VLMC models fit on elements of the same clan. The *between* represent the distance *between* VLMC models of different clans.



Supplementary Figure S8

VLMC distance and geographical overlap relation by coda type.

each bar represents a number of unique coda types. Note that almost all coda types have a negative correlation with geographical overlap, although only a small portion is significant. The p-values shown are with respect to the Pearson correlation, [Table S4](#)

Clan	K-S test	T-test	Effect Size
EC1	1.5×10^{-16}	1×10^{-24}	-2.4
EC2	0.0019	0.0006	-1.9
Four Plus	1.7×10^{-7}	6.2×10^{-6}	-0.67
Palindrome	6.7×10^{-17}	6.7×10^{-13}	-1.3
Plus One	6.9×10^{-21}	2.4×10^{-14}	-1.6
Regular	4.2×10^{-27}	7.2×10^{-28}	-0.76
Rapid Increasing	0.00016	0.0005	-0.72
Short	6.5×10^{-14}	9.9×10^{-18}	-0.67
Slow Increasing	1.6×10^{-9}	1.6×10^{-7}	-1.4

Supplementary Table S3

Distance distribution statistics.

Table with the p- values and the effect size corresponding to the comparison of the *within/between* distance distributions of coda samples from the Dominica clans (top) and the Pacific clans (bottom), under the Kolmogorov-Smirnov (K-S test) and the T-Test.

3.4.3. Stability under different resolutions

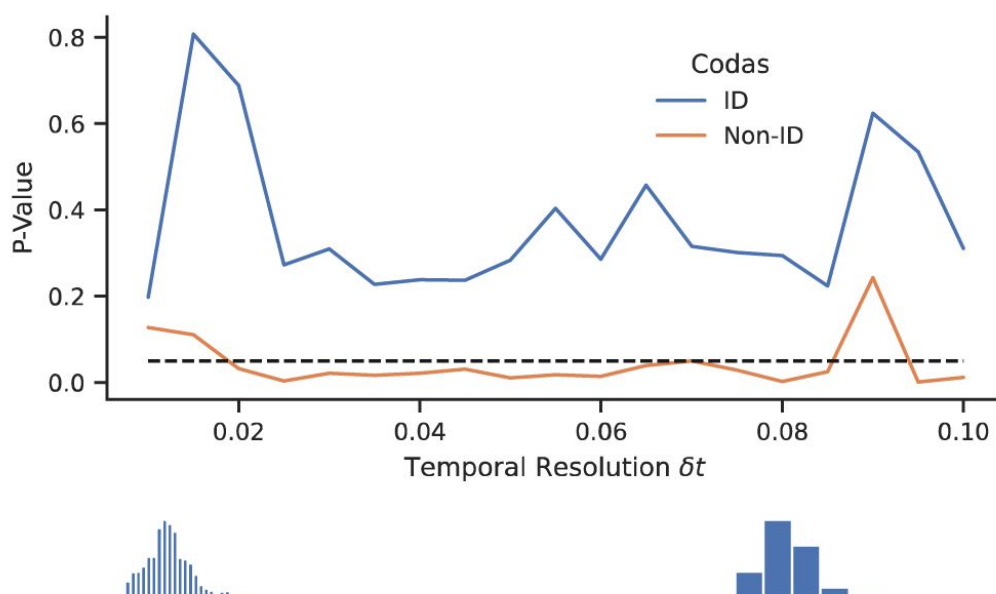
We also show that our results about the effect of sympatry on non-ID coda vocal styles hold for different values of the time resolution. That is, that the parameter preprocessing steps and the method parameters have little to no effect on the fundamental results of our approach. In **Fig. S9** we repeat the analysis of the main text. We compare the geographical overlap with the distance between the VLMC of the pacific clans on both non-ID and ID codas.

We observe that regardless of the time resolution used in the method (for discretizing the continuous ICIs into discrete ICIs), our results hold. That is, there is never a significant correlation between overlap and ID codas and that there is always a negative correlation between clan overlap and non-ID codas.

3.5 Dependence on Coda Type

In this section we provide results highlighting the lack of correlation between VLMC (subcode tree) similarity and coda type distribution. First, rhythmic variations on how each coda type is *constructed* are present and are indicative of the clan (**Fig. S11** and **Fig. S10**). For example, the way the clan EC1 vocalizes codas of type *8R* is significantly different from the clan EC2 (**Fig. S11**).

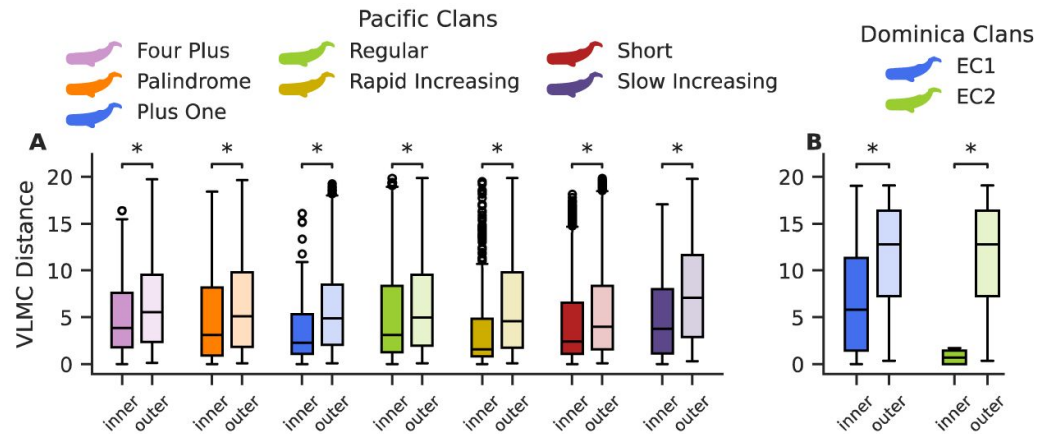
In fact one can fit a VLMC on each coda type and compare each VLMC (segmented by coda type and clan) between both elements of the same clan and elements of different clans. We observe that there is a statistically significant difference between the distances of VLMC from different clans and VLMC of the same clan (**Fig. S10**). This indicates that whales vocalize different coda types in a clan-distinctive manner. Which also point to an independence between vocal style and coda type distribution.



Supplementary Figure S9

Results are significant for different resolutions.

The dotted line represents the 0.05 confidence value. When varying the time resolution used for fitting the VLMCs we observe that our main results persist: ID codas are not related with geographical overlap while non-ID codas always are. The bottom row shows two illustrations of the bin sizes (and subsequently the number of different states) resulting from the preprocessing using 0.01 and 0.1 seconds time resolution respectively.



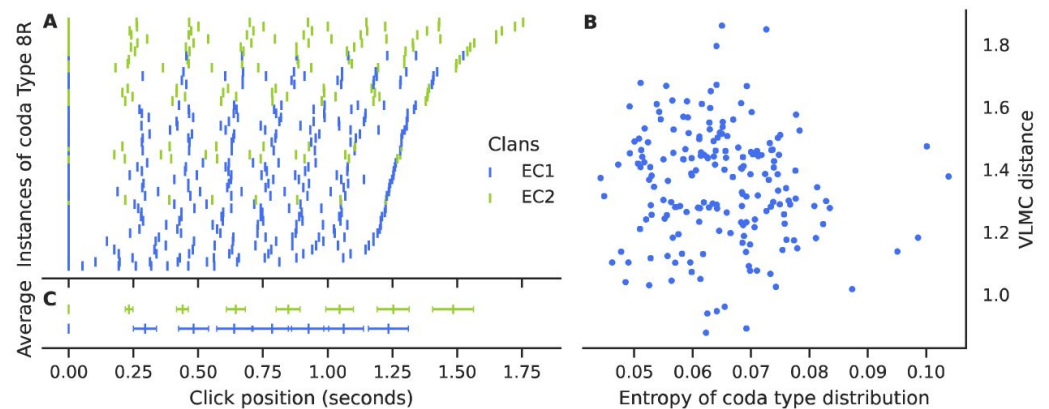
Supplementary Figure S10

Clans vocalize different coda types in a distinctive manner

Boxplots for the distances between VLMC's segmented by coda type and clan (Pacific **A** and Dominica **B**). We observe that VLMC fit on different coda types but from the same clan (inner) are more similar than when compared with across clans (outer).

3.5.1 Comparing Dendrograms

Using the distance between the VLMC trees it is possible to create a hierarchical plot of the coda samples. One can find it beneficial to compare our resulting hierarchical plot with the clan labels from [1] where the authors group the whales by coda type usage and divide the Pacific clans into the aforementioned 7 clans.



Supplementary Figure S11

A) and C) Rhythmic variations within coda types.

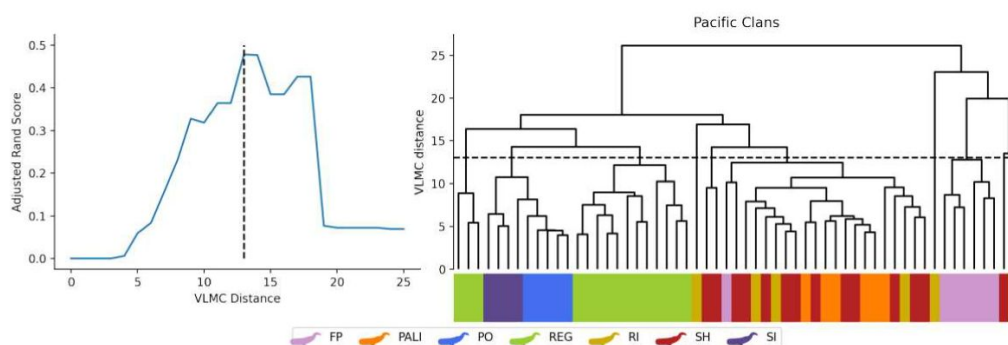
All codas of type 8R that were vocalized by whales of the Dominican clan EC1 (blue) and EC2 (green), along with the average position and variation over all vocalizations. **B) No correlation with coda type distribution.** Comparison between VLMC distance and KL-Divergence (entropy) between coda type distributions on the data used to fit the VLMC.

However, comparing a dendrogram with a realized set of labels is not trivial. On the other hand, an effective comparison of two sets of labels can be achieved using the *Adjusted Rand Score*, or other entropy based metrics. The Adjusted Rand Score has a value of 0.0 for random labeling (independent of number of clusters) and 1.0 for clusters that match perfectly. The lowest possible score is -0.5 for exceptionally disparate clusterings.

To obtain two sets of labels we progressively cut the dendrogram obtained by the VLMC and compared the set of labels with the clan labels from [1]. At each cut we calculate the adjusted rand score (results in Fig. S12). We observed a maximum value of 0.5. This reiterates not only the concordance with the pre-existing vocal usage clans but also emphasizes that vocal style is capturing new information at a different, lower, scale.

3.6 Confidence Interval on relation between non-id coda style and clan overlap

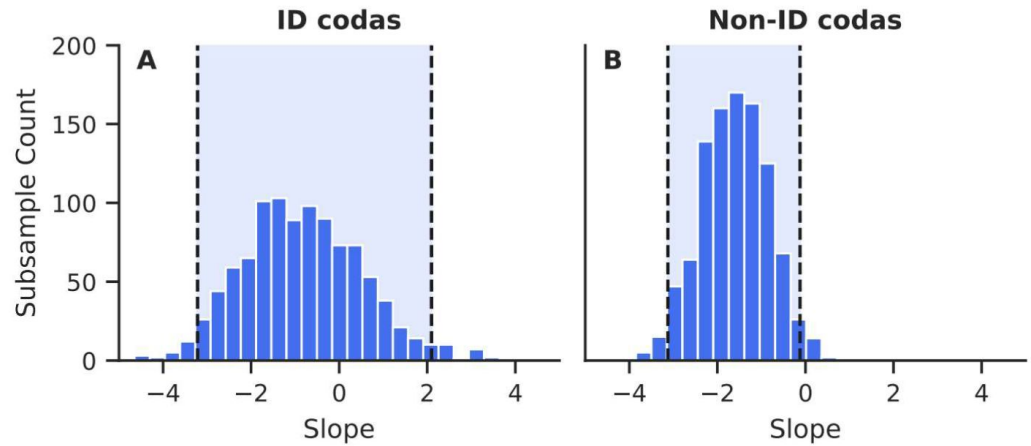
An interval confidence for the slope for the result in Figure 4 can be achieved by subsampling the data (1000 times) and running the same linear regression analysis on the subsampled data. From the resulting distribution of regression slopes we observe that the 95% confidence intervals for the non-ID scenario contains only values with negative slope, while on the ID case, the confidence interval contains both negative and positive values (Fig. S13).



Supplementary Figure S12

Adjusted Rand Score for VLMC clustering dendrogram

Comparison of the dendrogram obtained from the VLMC distance between the trees with the vocal clan labels. The values of the Adjusted Rand Score go from -0.5 to 1.



Supplementary Figure S13

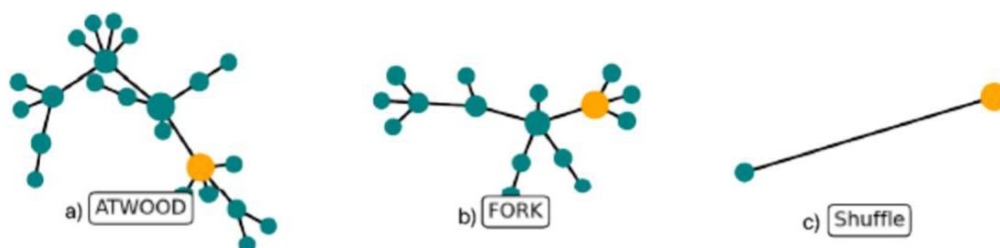
Bootstrapping regression results of Figure 4 [↗](#).

Resulting distribution of slope values for subsamples of the data for the ID coda case **A** and the non-ID case **B**. The shaded area represents the 95% confidence interval.

4 Classification of synthetic codas

The fixed-length Markov chains described above lack flexibility: a model with large memory h generalizes better but requires estimating an ever-increasing number of parameters. For this reason, we then used VLMCs, which combine the best of both worlds by determining the optimal memory needed for each transition individually. Essentially, we keep a transition probability with a longer memory $P(X_i | x_{j:k})$ only if it changes the distribution sufficiently compared to a short memory one $P(X_i | x_j)$.

A VLMC can be naturally visualized as a tree, where the concept of order arises from the fact that shorter memory contexts are subsequences of longer ones. In **Fig. S14A and B**, we show examples of two VLMCs computed from data from a single sperm whale each. The visual structure of these trees can be related to the actual information-theoretic structure of these sperm whales' communication. Indeed, the root node is represented in orange, and nodes that are depth h in the tree (that is h edges away from the root node) represent context (or sequences) of memory h . To verify that the structure we observe actually contains information, we need to compare it to the structure of a null model. To do this, we took the same ICI time series used to build the tree from **Fig. S14A**, and randomly shuffled its ICIs. This way, all temporal information is lost. This results in a tree that has no structure, as shown in **Fig. S14C**. The VLMC-indicated structure can thus be interpreted as coming from the sperm whale communication.

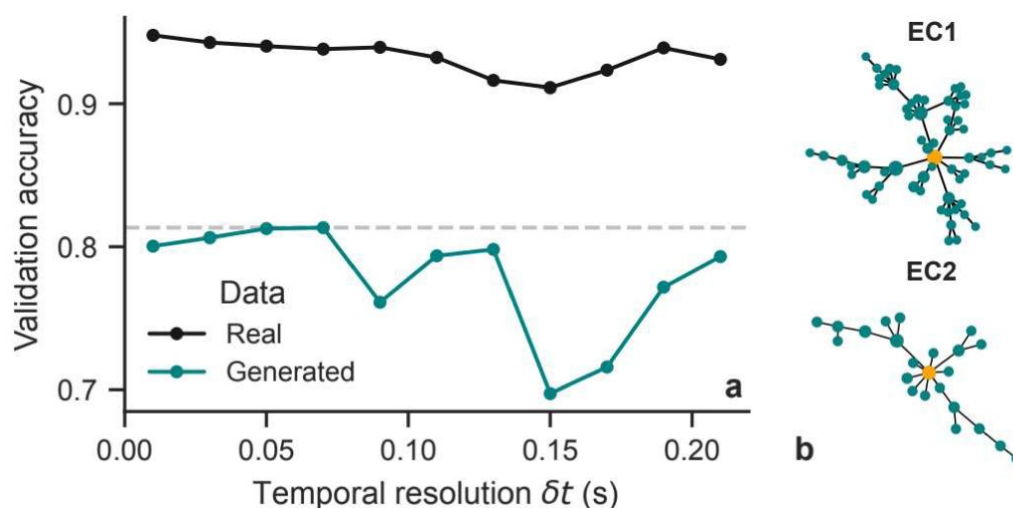


Supplementary Figure S14

VLMCs capture structure of sperm whale communication.

We show context trees associated with the variable length models built for two individual sperm whales: (a) *ATWOOD*, and (b) *FORK*. For comparison, we show (c) the corresponding tree after randomly shuffling ICIs from the same timeseries data used to build (a). Note how the shuffled version does not exhibit any structure. The orange node represents the root node, and the size of each node represents the number of occurrences of the associated context.

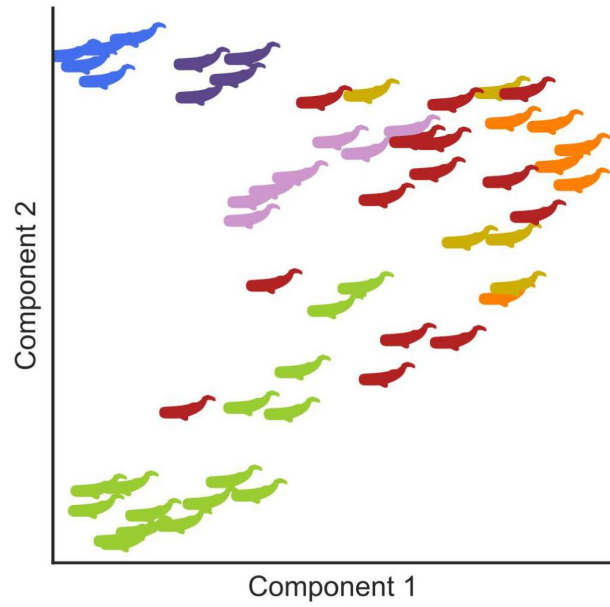
Having confirmed that our VLMCs capture some communication structure, we ask: What and how much structure does it capture? To answer this, we took advantage of two facts. First, the VLMCs can be used to generate new codas by generating sequences of states that correspond to ICIs. Indeed, like for any Markov model, we can start from the empty sequence, and start adding suffixes with probabilities defined by the model (see Methods). In other words, we can generate synthetic data. Second, in [10] the authors present an LSTM-based classifier capable of assigning a coda to a specific clan with over 90% accuracy. We trained it on the original ICI data used to build our trees, achieving similar accuracy as shown by the black curve in **Fig. S15**. To verify how much information our VLMCs capture, we used that trained classifier on the synthetic codas generated with our trees. Remarkably, it classified the generated data with between 70 and 80% accuracy, depending on the temporal resolution δt (blue in **Fig. S15**). The fairly small difference in accuracy between the real and synthetic data indicates that a large part of the communication structure captured by the classifier in the real data is also captured by our VLMC models.



Supplementary Figure S15

Deep learning classifier trained on real communication data generalises well to synthetic data generated by the VLMCs.

We show (a) the accuracy of the classifier on the real data (in black) and the synthetic data (in blue), as a function of the temporal resolution δt . The dashed grey curve highlights the maximum accuracy of on the generated data. The classifier's task was to identify to which of (b) two clans the data belonged to: EC1 or EC2.



Supplementary Figure S16

Vector Space embedding using UMAP

Embedding of all the Pacific Ocean clans according to their VLMC distance into two dimensions using UMAP.

Coda Type	Slope	Pearson-P	Spearman-P	Number of Clans
33	-0.353	0.864	0.787	3
34	-0.944	0.257	0.050	4
35	-6.022	0.031	0.032	5
42	-4.098	0.004	0.010	7
43	-0.615	0.596	0.573	4
45	-1.327	0.266	0.207	5
47	-5.808	0.628	0.700	3
48	9.818	0.258	0.257	3
49	10.481	0.371	0.538	3
51	-0.939	0.634	0.695	6
52	-1.663	0.002	0.021	7
53	-3.341	0.017	0.075	6
55	0.522	0.909	0.752	4
57	-3.480	0.410	0.511	4
58	-0.896	0.755	0.623	3
61	-1.512	0.074	0.006	7
64	-1.651	0.061	0.276	5
66	-2.482	0.138	0.062	7
67	-2.782	0.172	0.269	6
71	-1.994	0.870	0.518	3
72	-0.041	0.979	0.856	7
73	-1.334	0.337	0.329	3
76	-0.981	0.681	0.829	7
77	-4.093	0.141	0.093	5
79	-1.116	0.268	0.142	5
82	-0.847	0.265	0.266	3
83	0.908	0.578	0.284	6
85	-1.473	0.044	0.032	4
86	-0.089	0.845	0.600	3
88	2.831	0.000	0.000	7
91	-2.662	0.254	0.387	6
93	0.613	0.597	0.618	6
310	-4.118	0.320	0.193	4
311	-1.554	0.170	0.016	6
312	-1.275	0.325	0.069	6
313	-3.087	0.475	0.186	4
315	-1.189	0.593	0.700	3
411	0.489	0.711	0.795	4
412	-2.092	0.578	0.700	3
414	-2.269	0.012	0.006	7
511	-3.246	0.038	0.059	7
513	-1.806	0.174	0.107	7
514	-0.449	0.869	1.000	4
610	-1.794	0.145	0.123	7
612	5.633	0.486	0.618	3
613	-0.426	0.907	0.397	3
1011	-1.081	0.539	0.660	5
1012	-0.966	0.696	0.342	4

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Supplementary Table S4

Geographical overlap and subcode distance by coda type.

Results of comparing the geographical overlap and VLMC distance segmented by coda types. Negative correlations are highlighted as well as p-values that are below 0.05. “Number of Clans” represents how many clans were compared.

Clan	FP	RI	P	PO	R	S	SI
Four-Plus (FP)	1	0.500	0.385	0.308	0.885	0.962	0.308
Rapid Increasing (RI)	0.368	1	0.263	0.316	0.316	0.842	0.316
Palindrome (P)	0.800	0.733	1	0.733	0.733	0.800	0.800
Plus-One (PO)	1	1	1	1	1	1	1
Regular (R)	1	0.840	0.820	0.820	1	0.520	0.820
Short (S)	0.580	0.580	0.400	0.400	0.520	1	0.420
Slow Increasing (SI)	0.938	0.938	0.938	0.938	0.938	1	1

Supplementary Table S5

Clan spatial overlap values for Pacific Ocean sperm whale clans as calculated in [1]. The matrix gives the proportion of the row clan’s repertoires that were recorded within 1,000 km of at least one of the column clan’s repertoires. 1,000 km was chosen because it is the approximate annual home range of eastern tropical Pacific sperm whales (Whitehead, 2001; Whitehead et al., 2008). The matrix is asymmetric because a clan found in only one part of the Pacific Ocean might overlap completely with a clan that spans the Pacific Ocean, while the inverse is not true.

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Reviewer #2 (Public review):

Summary:

The current article presents a new type of analytical approach to the sequential organisation of whale song units.

Strengths:

The detailed description of the internal temporal structure of whale songs is something that has been thus far lacking.

Weaknesses:

The conceptual and terminological bases of the paper are problematical and hamper comparison with other taxa, including humans. According to signal theory, codas are indexical rather than symbolic. They signal an individual's group identity. Borrowing from humans and linguistics, coda inter-group variation represents a case of accents - phonologically different varieties of the same call - not dialects, confirming they are an index. This raises serious doubt about whether alleged "symbolism" and similarity between whale and human vocal behaviour is factual. The same applies to the difference between ICIs (inter-click interval) and IOIs (inter-onset interval). If the two are equivalent, variation in click duration needs to be shown so small that can be considered negligible. This raises serious doubt about whether the alleged variation in whale codas is indeed rhythmic in nature and prevents future efforts for comparison with the vocal capacities of other species. The scope and relevance of this paper for the broader field is limited.

<https://doi.org/10.7554/eLife.96362.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Summary:

This manuscript presents evidence of 'vocal style' in sperm whale vocal clans. Vocal style was defined as specific patterns in the way that rhythmic codas were produced, providing a fine-scale means of comparing coda variations. Vocal style effectively distinguished clans similar to the way in which vocal repertoires are typically employed. For non-identity codas, vocal style was found to be more similar among clans with more geographic overlap. This suggests the presence of social transmission across sympatric clans while maintaining clan vocal identity.

Strengths:

This is a well-executed study that contributes exciting new insights into cultural vocal learning in sperm whales. The methodology is sound and appropriate for the research question, building on previous work and ground-truthing much of their theories. The use of the Dominica dataset to validate their method lends strength to the concept of vocal style and its application more broadly to the Pacific dataset. The results are framed well in the context of previous works and clearly explain what novel insights the results provide to the current understanding of sperm whale vocal clans. The discussion does an overall great job of outlining why horizontal social learning is the best explanation for the results found.

Weaknesses:

The primary issues with the manuscript are in the technical nature of the writing and a lack of clarity at times with certain terminology. For example, several tree figures are presented and 'distance' between trees is key to the results, yet 'distance' is not clearly defined in a way for someone unfamiliar with Markov chains to understand. However, these are issues that can easily be dealt with through minor revisions with a view towards making the manuscript more accessible to a general audience.

I also feel that the discussion could focus a bit more on the broader implications - specifically what the developed methods and results might imply about cultural transmission in other species. This is specifically mentioned in the abstract but not really delved into in detail during the discussion.

We are grateful for the Reviewer's recognition of the study's contributions to understanding cultural vocal learning in sperm whales. In response to the concerns regarding clarity and accessibility, we have revised the manuscript to improve the definition of key concepts, such as the notion of "distance" between subcode trees. This adjustment ensures clarity for readers unfamiliar with the technical details of Markov chains. Additionally, we have expanded the discussion to highlight broader implications of our findings, particularly their relevance to understanding cultural transmission in other species, as suggested.

Reviewer #2 (Public review):

Summary:

The current article presents a new type of analytical approach to the sequential organisation of whale coda units.

Strengths:

The detailed description of the internal temporal structure of whale codas is something that has been thus far lacking.

Weaknesses:

It is unclear how the insight gained from these analyses differs or adds to the voluminous available literature on how codas varies between whale groups and populations. It provides new details, but what new aspects have been learned, or what features of variation seem to be only revealed by this new approach? The theoretical basis and concepts of the paper are problematical and indeed, hamper potentially the insights into whale communication that the methods could offer. Some aspects of the results are also overstated.

We appreciate the Reviewer's acknowledgment of the novelty in describing the internal temporal structure of whale codas. Regarding the concern about the unique contributions of this approach, we have further emphasized in the revised manuscript how our methodology reveals previously uncharacterized dimensions of coda structure. Specifically, our work highlights how non-identity codas, which have received limited attention, play a significant role in inter-clan acoustic interactions. By leveraging Variable Length Markov Chains, we provide a nuanced understanding of coda subunits that complements existing studies and demonstrates the value of this analytical approach.

Reviewer #3 (Public review):

Summary:

The study presented by Leitao et al., represents an important advancement in comprehending the social learning processes of sperm whales across various communicative and socio-cultural contexts. The authors introduce the concept of "vocal style" as an addition to the previously established notion of "vocal repertoire," thereby enhancing our understanding of sperm whale vocal identity.

Strengths:

A key finding of this research is the correlation between the similarity of clan vocal styles for non-ID codas and spatial overlap (while no change occurs for ID codas), suggesting that social learning plays a crucial role in shaping symbolic cultural boundaries among sperm whale populations. This work holds great appeal for researchers interested in animal cultures and communication. It is poised to attract a broad audience, including scholars studying animal communication and social learning processes across diverse species, particularly cetaceans.

Weaknesses:

In terms of terminology, while the authors use the term "saying" to describe whale vocalizations, it may be more conservative to employ terms like "vocalize" or "whale speech" throughout the manuscript. This approach aligns with the distinction between human speech and other forms of animal communication, as outlined in prior research (Hockett, 1960; Cheney & Seyfarth, 1998; Hauser et al., 2002; Pinker & Jackendoff, 2005; Tomasello, 2010).

We thank the Reviewer for recognizing the importance of our findings and their appeal to broader audiences interested in animal cultures and communication. In response to the suggestion regarding terminology, we have adopted a more conservative language to align with distinctions between human and non-human communication systems. For example, terms like "vocalize" and "vocal repertoire" are used in place of anthropomorphic terms such as "saying". This ensures consistency with established conventions while maintaining clarity for a broad readership.

Reviewer #1 (Recommendations):

Comment 1

Lines 11-13: As mentioned above, the implications for comparing communication systems and cultural transmission in other species isn't really discussed much and I think it's a really interesting component of the study's broader implications.

Thank you for the comment.

Action - We added a few more sentences to the discussion regarding this.

Comment 2

Figure 1: More information on the figure of these trees would help. What do the connecting lines represent? What do the plain black dots and the black dot with the white dot represent? Especially since the "distance between trees" is a key result, it's important that someone unfamiliar with Markov chains can understand the basics of how this is calculated and what it represents. It is explained in the methods, but a brief explanation here would make the results and the figure a lot clearer since the methods are the last section of the manuscript.

These were omitted as we believed that attempting to introduce the mathematical structure and the methodology to compare two instances, in a figure caption, would have caused more ambiguity than necessary.

Action - Added an informal introduction to these concepts on the figure caption. Also added a pointer to the Supplementary Materials.

Comment 3

Table 1: A definition of dICIs should be included here.

Added the definition of discrete ICI to the table.

Comment 4

Figure 2: The placement of the figures is a bit confusing because they are quite far from the text that references them.

We thank the reviewer for pointing this out, we tried to edit the manuscript to improve this issue, but this part of the editing is more within the journal's powers than our own.

Action - Moved images closer to the corresponding text in manuscript.

Comment 5

Line 117: Probabilistic distance needs to be briefly explained earlier when you first mention distance (see Lines 11-13 comments).

Action - Clarifications added in the caption of figure 1. as per comment on Lines 11-13

Comment 6

Figure 4: Is order considered in these pairwise comparisons? It looks like there are two dots for each pairwise comparison. Additionally, why is the overlap different in these two comparisons? For example, short:four-plus has an overlap of 0.6, while four-plus:short has an overlap of 0.95.

The x-axis of the plots in Figure 4 is geographical clan overlap. This is calculated as per (Hersh et al., 2022) and is described in our Methods (see "Measuring clan overlap" section). Given two clans—for example, the Four-Plus and the Short clan—spatial overlap is calculated twice: as the proportion of the Four-Plus clan's repertoires that were recorded within 1,000 km of at least one of the Short clan's repertoires, and as the proportion of the Short clan's repertoires that were recorded within 1,000 km of at least one of the Four-Plus clan's repertoires.

Order is important in these pairwise comparisons and generates an asymmetric matrix because the clans have different spatial extents. A clan found in only one small region might overlap completely with a clan that spans the Pacific Ocean, while the opposite is not true. For example, the Short clan spans the Pacific Ocean while the Four-Plus clan has been documented over a smaller area (but that smaller area overlaps extensively with the Short clan range). That is why the value is smaller (0.6) when considering how much of the Short clan's range is shared with the Four-Plus clan, and larger (0.95) when considering how much of the Four-Plus clan's range is shared with the Short clan.

Action - We have now added a reference to that section of the Methods in our Figure 4 caption and include the clan spatial overlap matrix as a supplemental table (Table S5).

Comment 7

Figure 4: I think the reference should be Hersh et al. [11].

Thank you for catching this.

Action - Reference corrected

Comment 8

Line 227: What aspect of your analysis looked at how often codas were produced? You mention coda frequency, but it is unclear how this was incorporated into your analysis. If this is included in the methods, the language is a bit too technical to easily parse it out.

Indeed here we are referencing the results of the paper mentioned in the previous line. We do not look at coda production frequency.

Action - Added citation to paper that actually performs this analysis.

Comment 9

Lines 253-255: I think you could dig into this a little more, as "there is currently no evidence" is not the most convincing argument that something is not a driver. Perhaps expanding on the latter sentence that clans are recognizable across oceans basins would be helpful. Does this suggest that clans with similar geographic overlap experience diverse environmental conditions across ocean basins? If so, this might better strengthen your argument against environmental drivers.

Thank you for pointing this out. We feel that the next sentence highlights that clans are recognizable across environmental variation from one side to the other of the ocean basin, which supports the inductive reasoning that codas do not vary systematically with environment. However, we have edited these sentences for clarity.

Comment 10

Lines 311-314: It would also be interesting to look at vocal style across non-ID coda types. Are some more similar to each other across clans than others? Perhaps vocal style can further distinguish types of non-ID codas.

In supplementary Materials 3.4.2 and 3.5 we highlight our results when the codas are separated by coda type summarized in Table S4. We do compare the vocal style across non-ID coda types across clans and within the same clan. The results however are aggregated to highlight the differences in style between the clans and a coda type-only comparison is not shown.

Comment 11

Lines 390-392: I'm assuming this is why pairwise comparisons were directional (i.e., there was both an A:B and a B:A comparison)? Can you speak to why A:B and B:A comparisons can have such different overlap values?

Given two clans—for example, the Four-Plus and the Short clan—spatial overlap is calculated twice: as the proportion of the Four-Plus clan's repertoires that were recorded within 1,000 km of at least one of the Short clan's repertoires, and as the proportion of the Short clan's repertoires that were recorded within 1,000 km of at least one of the Four-Plus clan's repertoires.

Order is important in these pairwise comparisons and generates an asymmetric matrix because the clans have different spatial extents. A clan found in only one small region might overlap completely with a clan that spans the Pacific Ocean, while the opposite is not true. For example, the Short clan spans the Pacific Ocean while the Four-Plus clan has been documented over a smaller area (but that smaller area overlaps extensively with the Short clan range). That is why the value is smaller (0.6) when considering how much of the Short clan's range is shared with the Four-Plus clan, and larger (0.95) when considering how much of the Four-Plus clan's range is shared with the Short clan.

Action - We now include the clan spatial overlap matrix as a supplemental table (Table S5).

Comment 13

Line 56: Can you briefly explain what memory means in the context of Markov chains?

We provide an explanation of the meaning of memory in the Methods section on "Variable length Markov Chains". Briefly, the memory in this case means how many states in the past of the Markov chain's current state are required to predict the next transition of the chain itself. Standard Markov chains "look" back only one time step, while k -th order Markov chains look back k steps. In our case, there was no reason to assume that the memory required to predict different sequences of states (interclick intervals) should be the same across all sequences, and thus we adopted the formalism of variable length Markov chains, that allow for different levels of memory across the system.

Comment 14

Supplementary Figure S3: Like in the main manuscript, briefly explain or remind us what the blank nodes and the yellow nodes are.

Action - Clarified that the orange node represents the root of the tree in the figures.

Comment 15

Supplementary Figure S7: Put the letters before the dataset name.

Action - Done.

Comment 16

Supplementary Figure S10: Unclear what 'inner vs outer' means.

One specifies comparisons across clans (outer) and the other within the same clan (inner)

Action - Added clarification on the caption of Figure S10

Comment 17

Supplementary Figure S14: Include a-c labels in the figure itself.

Action - Labels added to figure

Comment 18

Supplementary Figure S14: The information about the nodes is what needs to be included earlier and in the main body when discussing the trees.

Action - Added the explanation earlier in the text and in the main body

Reviewer #2 (Recommendations):

Comment 19

Line 22: "Symbolic" and "Arbitrary" are not synonyms. Please see the comment above.

We agree. Here, we make the point that the evolution of symbolic markers of group identity can be explained from what are initially arbitrary, and meaningless, signals (see [L1, L2]). Our point being that any vocalization, any coda, could have become selected for as an identity coda, and to become symbolic, and evolve to play a key role in cultural group formation and in-group favoritism because they enable a community of individuals to solve the problem of with whom to collaborate. The specific coda itself does not affect collaborative pay offs, but group specific differences in behavior can, as such the coda is arguably symbolic; as it is observable and recognizable, and can serve as a means for social assortment even when the behavioural differences are not. This can explain the means by which the social segregation which is observed among behaviorally distinct clans of sperm whales. However, in this manuscript, we do not extend this discussion of existing literature and have attempted to concisely describe this in a couple of lines, which clearly do a disservice to the large body of literature on the evolution of symbolic markers and human ethnic groups. We have added some citations to this section so that the reader may follow up should they disagree with our brief introductory statements.

Action - Added citations and pointers to the literature.

Comment 20

Line 24: The authors' terminology around "markers", "arbitrary", "symbolic" is unnecessarily confusing and mystifying, giving the impression these terms are interchangeable. They are not. These terms are an integral and long-established part of key definitions in signal theory. Term use should be followed accordingly. The observation that whale vocal signals vary per population does not necessarily mean that they function as a social tag. The word "dog" varies per population but its use relates to an animal, not the population that utters the word. "Dog" is not "symbolic" of England, English-speaking populations or the English language. Furthermore, the function of whale vocal signals is extremely challenging to determine. In the best conditions, researchers can pin the signal's context, this is distinct from signal's function and further even for the signal's meaning. How exactly the authors determine that whale vocal signals are arbitrary is, thus, perplexing given that this would require a detailed description and understanding of who is producing the song, when, towards whom, and how the receivers react, none of which the authors have and without which no claim on the signals' function can be made. This terminological laxness and the sensu latu in extremis to various terms in an unjustified, unnecessary and unhelpful.

We use these terms as established in Hersh et al 2022 and the works leading up to it over the last 20 years in the study of sperm whales. These are often derived from definitions by Boyd and Richerson's work on culture in humans and animals along with evolution of symbolic markers both in theory and in humans. We agree with the reviewer that these are difficult to establish in non-humans, whales or otherwise, but feel strongly that the accumulating evidence provides strong support for the function of these signals as symbolic markers of cultural groups, and that they likely evolved from initially arbitrary calls which were a part of the vocal repertoire (similar to the process and selective environment in Efferson et al. [L1] and McElreath et al. [L2]). We feel that we do not use these terms interchangeably here, and have inherited their use from definitions from anthropology. The work presented here uses terminology built across two decades of work in cetacean, and sperm whale, culture. And do not feel that these terms should be omitted here.

Comment 21

Lines 21-27: Overly broad and hazy paragraph.

We hope the replies above and our changes satisfy this comment and clarify the text.

Comment 22

Figure 1 legend: What are "memory structures"? Unjustified descriptor.

The phrase was chosen to make draw some intuition on the variation of context length in variable length markov models.

Action - Re-worded from memory structures to statistical properties

Comment 23

Line 30: Omit "finite".

Action - Omitted.

Comment 24

Line 31: Please define and distinguish "rhythm" and "tempo". Also see comment above, rhythm and tempo definitions require the use of IOIs.

We disagree with the reviewer's claims here. In our research specifically, and for sperm whale research generally, coda inter-click intervals (ICIs) are calculated as the time between the start of the first click and the start of the subsequent click. This makes ICIs identical to inter-onset intervals (IOIs) under all definitions we are aware of. For example, Burchardt and Knornschild [L3] define IOIs as such: "In a sequence of acoustic signals, the time span between the start of an element and the next element, comprising the element duration and the following gap duration". We now include a sentence making this point.

Regardless, we disagree on a more fundamental level with the statement that unless researchers quantify inter-onset intervals (IOIs), they cannot make any claims about rhythm. There are many studies that investigate rhythmic aspects of human and animal vocalizations without using IOIs [L4-L7]. If the duration of sound elements of interest is relatively constant (as is the case for sperm whale clicks), then rhythm analyses can still be meaningfully conducted on inter-call intervals (the silent intervals between calls).

For sperm whales, coda rhythm is defined by the relative ICIs standardized by their total duration. These can be clustered into discrete, defined rhythm types based on characteristic

ICI patterns. Coda tempo is relative to the total duration of the coda itself. This can also be clustered into discrete tempo types across all coda durations as well (see [L8]).

Action - We added a sentence specifying that in this case we can use both ICIs and IOIs because of the standardized length of a single click.

Comment 25

Line 36: *Are there non-vocalized codas to require the disambiguation here?*

No, we have omitted for clarity.

Comment 26

Line 44: *"Higher" than which other social group class?*

Sperm whales live in a multi-level social organization. Clans are a "higher" level of social organization than the social "units" which we define in line 40. Clans are made up of all units which share similar production repertoire of codas.

Action - We have added 'above social units' on line 44 to make this clear.

Comment 27

Line 47: *The use of "symbolic" continues to be enigmatic, even if authors are taking in this classification from other researchers. In signal theory (semiotics), not all biomarkers are necessarily symbols. I advise the authors to avoid the use of the term colloquially and instead adopt the definition used in the research field within which the study falls in.*

There is ample examples of the use of "symbolic" when referring to markers of in-group membership both in human and non-human cultures. Our choice to use the term "symbolic" is based on a previous study [L9] that found quantitative evidence that sperm whale identity codas function as symbolic markers of cultural identity, at least for Pacific Ocean clans. The full reasoning behind why the authors used the term "symbolic markers" is given in that paper, but briefly, they found evidence that identity coda usage becomes more distinct as clan overlap increases, while non-identity coda usage does not change. This matches theoretical and empirical work on human symbolic markers [L1, L2, L10, L11].

Action - We retain the use of the term here, as defined in the works cited, and based on its prior usage in the study of both human and non-human cultures.

Comment 28

Line 50: *This statement is not technically accurate. The use of a signal as a marker by individuals can only be determined by how individuals "interpret" and react to that signal - e.g., via playback experiments - it cannot be determined by how different populations use and produce the signals.*

We respectfully disagree. While we agree that the optimal situation would be that of playback, the contextual use can provide insight into the functional use of signals; as can expected patterns of use and variation, as was tested in the papers we cite. However, this argument is not the scope nor the synthesis of this paper. These statements are supported by existing published works, as cited, and we encourage the reviewer to take exception with those papers.

Comment 29

Line 69: "Meaningful speech characteristics"??? These terms do not logically or technically follow the previous statement. Why not stay faithful to the results and state that the method used seems to be valid and reliable because it confirms former studies and methods?

Action - Reworded to better underline the method's results with previous studies

Comment 30

Lines 72-74: This statement doesn't seem to accurately capture/explain/resume the difference between ID and non-ID codas.

We are not sure what the reviewer is referring to in this case. The sentence in this case was meant to explain the different relations that ID/non-ID codas have with clan sympatry.

Comment 31

Line 75: The information provided in the few previous sentences does not allow the reader to understand why these results support the notion that cultural transmission and social learning occurs between clans.

We conclude out introduction with a brief summary of our overall findings, which we then use the rest of the manuscript to support these statements.

Comment 32

Table 1: So far, the authors refer to their analyses as capturing the "rhythm" of whale clicks. Consequently, it is not readily clear at this point why the authors rely on "ICIs" (inter click intervals) instead of the "universal" measure used across taxa to capture the rhythm of signal sequences - IOIs (inter onset intervals). If ICIs are the same measure as IOIs, why not use the common term, instead of creating a new term name? Alternatively, if ICIs are not equivalent to IOIs, then arguably the analyses do not capture the "rhythm" of whale clicks, as claimed by the authors. Any rhythmic claim will need to be based on IOI measures. In animal behaviour, stereotyped is primarily used to describe pathological, dysfunctional behaviour. I suggest the use of other adjective, such as "regular", "repetitive", "recurring", "predictable". Another deviation from typical terminology: "usage frequency" -¿ "production rate". Why is a clan a "higher-order" level of social organization? This requires explanation, at least a mention, of what are the "lower-order" levels. To the non-expert reader, there is a logical circularity/gap here: Clans are said to produce clan-specific codas, and then, it is said that codas are used to delineate clans. Either one deduces, or one infers, but not both. This raises the question, are clans confirmed by any other means than codas?

We are not creating a "new term name": inter-click interval (ICI) is the standard terminology used in odontocete (toothed whale) research. We take the reviewer's point that some readers will not be coming to our paper with that background, however, and now explicitly point out that ICI is synonymous with IOI for sperm whales. Please see our response to your earlier comment for more on this point.

Comment 33

Line 92: Unclear term, "sub-sequence". Fig. 1B doesn't seem to readily help disambiguate the meaning of the term.

In fact reference to Fig. 1B is misplaced as it does not refer to the text. A sub-sequence is simply a contiguous subset of a coda, a subset of it.

Action - Removed ambiguous reference to Fig. 1B

Comment 34

Line 94: How does the use of "sequence" compare here with "sub-sequence" above?

In fact its the same situation although the previous comment highlighted a source of ambiguity.

Action - Reworded the sentence to be less confusing.

Comment 35

Line 95: Signal sequences don't "contain" memory, they require memory for processing.

Action - Rephrased from "sequences contain memory" to "states depend on previous sequences of varying length".

Comment 36

Lines 95-97: The analogy with human language seems forced, combinatorics in any given species are expected to entail different transitions between unit/unit-sequences.

Thank you for the comment. Indeed, the purpose of the analogy is to illustrate how variable length Markov Chains work (which have been shown to be good at discerning even accents of the same language). We used human language as an analogy to provide the readers' with a more intuitive understanding of the results.

Action - Revised paragraph to read: "Despite we do not have direct evidence of unitary blocks in sperm whale communication, on can imagine this effect similarly to what happens with words (e.g., a word beginning with "re" can continue in more ways than one starting with "zy")."

Comment 37

Line 97: Unclear which possibility is this.

Action - Made the wording clearer.

Comment 38

Line 99: Invocation of memory, although common in the use of Markov chains, is inadequate here given that the research did not study how individuals perceived or processed click sequences, only how individual produced click sequences. If the authors are referring to the cognitive load imposed by producing clicks sequences, terms such as "sequence planning" will be more accurate.

Here, we use the term "fixed-memory" in relation to the definition of a variable length Markov model. We feel that, in this section of the manuscript, the context is clear that it is a mathematical definition and in no way invokes the biological idea of memory or cognition. It is rather standard to use memory to describe the order of Markov chains. Swapping words in the definition of mathematical objects when the context is clear seems to cause unnecessary ambiguity.

Action - We clarified this in the manuscript (see comments above).

Reviewer #3 (Recommendations):

Comment 39

Line 16: Add "broadly defined" as there are many other more restricted definitions (see for example Tomasello 1999; 2009). Tomasello M (1999) The cultural origins of human cognition. Harvard University Press, Cambridge Tomasello M (2009) The question of chimpanzee culture, plus postscript (chimpanzee culture 2009). In: Laland KN, Galef BG (eds) The question of animal culture. Harvard University Press, Cambridge, pp 198-221.

Thanks for the clarification.

Action - We added the term "broadly" and added the last reference.

Comment 40

Line 22: Is all stable social learned behavior that becomes idiosyncratic and "distinguishable" considered symbolic markers? If not, consider adding "potentially."

No, but the evolution of cultural groups with differing behavior can reorganize the selective environment in such a way that it can favour an in-group bias that was not initially advantageous to individuals and lead to a preference towards others who share an overt symbolic marker that initially had no meaning and a random frequency in both populations. That is to say, even randomly assigned trivial groups can evolve arbitrary symbolic markers through in-group favouritism once behavioural differences exist even in the absence of any history of rivalry, conflict, or competition between groups. See for example [L1, L2].

Comment 41

Table 1: Identity codas are defined as a "Subset of coda types most frequently used by a sperm whale clan; canonically used to define vocal clans." Therefore, I infer that an identity coda is not exclusively used by a specific clan and may be utilized by other clans, albeit less frequently. If this is the case, what criteria determine the frequency of usage for a coda to be categorized as an identity or non-identity coda? Does the criteria used to differentiate between ID and non-ID codas reflect the observed differences in micro changes between the two and within clans?

The methods for this categorization are defined, discussed, and justified in previous work in [L9, L12]. We feel its outside the scope of this paper to review these details here in this manuscript. However, the differences between vocal styles discussed here and the frequency production repertoires which allow for the definition of identity codas are on different scales. The differences between identity and non-identity codas are not the observed differences in vocal style reported here.

Comment 42

Table 1: The definition of vocal style states that it "Encodes the rhythmic variations within codas." However, if rhythm changes, does the type of coda change as well? Typically, in musical terms, the component that maintains the structure of a rhythm is "tempo," not "rhythm." How much microvariation is acceptable to maintain the same rhythm, and when do these variations constitute a new rhythm?

Thank you for raising this important point about the relationship between rhythmic variations and coda categorization. In our definition, "vocal style" refers to subtle, micro-level variations in the rhythmic structure of codas that do not alter their overarching categorical identity. These microvariations are akin to "tempo" changes in musical terms, which can modify the expression of a rhythm without fundamentally altering its structure.

The threshold at which microvariations constitute a new rhythm, and thus a new coda type, remains an open question and is a limitation of current analytical approaches. In our study, we used established classification methods to group codas into types, treating variations within these groups as part of the same rhythm. Future work could refine these thresholds to better distinguish between meaningful rhythmic variation and the emergence of new coda types.

Comment 43

Table 1: Change "say" to "vocalize" (similarly as used in line 273 for humpback whales "vocalizations").

Thanks.

Action - Done.

Comment 44

Lines 33-35 and Figure 1-C: Can a lay listener discern the microvariations within each coda type by ear? Consider including sound samples of individual rhythmic microvariations for the same coda type pattern (e.g., Four plus, Palindrome, Plus One, Regular) to provide readers/listeners with an impression of their detectability. If authors considered too much or redundant Supplemental material at least give a sound sample for each the 4 subcodas modeled structures examples of 4R2 coda variations depicted in Figure 1-C so the reader can have an acoustic impression of them.

We do not think that human listeners would be able to all of the variation detected here. However, this does not mean that it is not important variation for the whales. Human observers being able to classify call variation aurally shouldn't be seen as a bar representing important biological variation for non-human species, given that their hearing and vocal production systems have evolved independently. Importantly, 'Four Plus', 'Palindrome', etc are names of Clans; sympatric, but socially segregated, communities of whale families, which share a distinct vocal dialect of coda types. These clans each have distinguishable coda dialects made up of dozens of coda types (and delineated based on identity codas), these are not names/categorical coda types themselves.

Action - We now provide audio samples of all coda types listed in Figure 1B in the paper's Github repository.

Comment 45

Line 69: As stated above, it may be confusing to refer to it as "speech." I suggest adding something like: "Our method does capture one essential characteristic of human speech: phonology." Reply 45.—Thank you for drawing our attention to this.

Action - We removed the word "speech" from the manuscript, using "communication" and/or "vocalization" depending on the context.

Comment 46

Line 111-112: Consider adding a sound sample of the variation of the 4R2 coda type that can be vocalized as BCC but also as CBB as supplementary data.

What the reviewer has correctly observed is that the traditional categorical coda type 'names' do not capture the variation within a type by rhythm nor by tempo.

Action - We have added samples of all coda types listed in Figure 1B in the paper's Github repo.

Comment 47

Figure 3: Include a sound sample for each of the 7 coda types in Figure 1B ("specific vocal repertoires") to illustrate the set of coda types used and their associated usage frequencies, or at least for each of the 7 coda types in Figure 3 and tables S1 and S2.

Sperm whales in the Eastern Caribbean produce dozens of rhythm types across at least five categorical tempo types [L8, L13]. The coda types represented in Figure 1B do not demonstrate all the variability inherent in the sperm whales' vocal dialect. Importantly, Figure 3, as well as table S1 and S2, refer to clan-level dialects not specific individual coda types.

Action - We added sound samples for each coda rhythm type listed in Figure 1B to the Github repository.

Comment 48

Lines 184-190: It is unclear what human analogy term is used for ID codas. This needs clarification.

We are not making an analogy in humans for the role of ID vs non-ID codas, but only providing the example of accents as changes in vocalization (style) without a change in the actual words used (repertoire).

Action - We tried to make it clearer in the manuscript.

Comment 49

Line 190: Change "whale speech" to "whale vocalizations."

Thanks.

Action - Done.

Comment 50

Figure 4: Correct citation number Hersh "10" to Hersh "11."

Thanks.

Action - Fixed the reference.

Comment 51

Lines 224-232: Clarify whether the reference to how spatial overlap affects the frequency of ID codas refers to shared ID codas between clans or the production frequency of each coda within the total repertoire of codas.

The similarity between ID coda repertoires we are referring to there is based on the ID codas of both clans.

More details on the comparison can be found in [L9].

Action - We added a sentence explaining the comparison is made using the joint set of ID codas.

Comment 52

Lines 240-241: What are non-ID codas vocal cues for?

Non-ID codas likely serve as flexible, context-dependent signals that facilitate group coordination, convey environmental or social context, and promote social learning, especially in mixed-clan or overlapping habitats. Their variability suggests multifunctional roles shaped by ecological and social pressures.

Comment 53

Lines 267-268: It's unclear whether non-ID coda vocal styles are genetically inherited or not, as argued in lines 257-258.

We did not intend to argue that non-ID coda vocal styles are genetically inherited. Instead, we aimed to present a hypothetical consideration: if non-ID coda vocal styles were genetically inherited, one would expect a direct correlation between vocal style similarity and genetic relatedness. This hypothetical framework was introduced to strengthen our argument that the observed patterns are unlikely to be explained by genetic inheritance, as such correlations have not been observed. While we acknowledge that we lack definitive proof to rule out genetic influences entirely, the evidence available strongly suggests that social learning, rather than genetic transmission, is the more plausible mechanism.

Action - Clarified in manuscript.

Comment 54

Line 277: Can males mate with females from different clans?

Yes, genetic evidence shows that males may even switch ocean basins.

Action - We have clarified that we mean the female members of units from different clans have only rarely been observed to interact at sea between clans.

Comment 55

Lines 287-292: Consider discussing the difference between controlled/voluntary and automatic/involuntary imitation and their implications for cultural selection and social learning (see Heyes 2011; 2012). Heyes, C. (2011). Automatic imitation. Psychological bulletin, 137(3), 463. Heyes, C. (2012). What's social about social learning?. Journal of comparative psychology, 126(2), 193.

Thank you for your insightful comment regarding this. The distinction between controlled/voluntary and automatic/involuntary imitation, as highlighted by Heyes [L14, L15], provides a potentially valuable framework for interpreting social learning mechanisms in sperm whales. Automatic imitation refers to reflexive, often unconscious mimicry driven by perceptual or motor coupling, while controlled imitation involves deliberate and goal-

directed efforts to replicate behaviors. Both forms likely play complementary roles in the cultural transmission observed in sperm whales.

This dual-process perspective highlights the potential for cultural selection to act at different levels. Automatic imitation may drive convergence in shared environments, promoting acoustic homogeneity and facilitating inter-clan communication. In contrast, controlled imitation ensures the preservation of clan-specific vocal traditions, maintaining cultural diversity. This interplay between automatic and controlled processes could reflect a balancing act between cultural assimilation and differentiation, underscoring the adaptive value of these mechanisms in dynamic social and ecological contexts.

Action - We have incorporated a short discussion of this distinction and its implications for our findings in the Discussion. Additionally, we have cited [L14, L15] to provide theoretical grounding for this interpretation.

Comment 56

Methods: Consider integrating the paragraph from lines 319-321 into lines 28-35 and eliminate redundant information.

Thanks.

Action - We implemented the suggestion, removing the first paragraph of the Dataset description and integrating the information when we introduce the concepts of codas and clicks.

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[L2] R. McElreath, R. Boyd, and P. Richerson, *Curr. Anthropol.* 44, 122 (2003).

[L3] L. S. Burchardt and M. Knornschild, *PLoS Computational Biology* 16, e1007755 (2020).

[L4] A. Ravignani and K. de Reus, *Evolutionary Bioinformatics* 15, 1176934318823558 (2019).

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[L6] D. Gerhard, *Canadian Acoustics* 31, 22 (2003).

[L7] N. Mathevon, C. Casey, C. Reichmuth, and I. Charrier, *Current Biology* 27, 2352 (2017).

[L8] P. Sharma, S. Gero, R. Payne, D. F. Gruber, D. Rus, A. Torralba, and J. Andreas, *Nature Communications* 15, 3617 (2024).

[L9] T. A. Hersh, S. Gero, L. Rendell, M. Cantor, L. Weilgart, M. Amano, S. M. Dawson, E. Slooten, C. M. Johnson, I. Kerr, et al., *Proc. Natl. Acad. Sci.* 119, e2201692119 (2022).

[L10] R. Boyd and P. J. Richerson, *Cult Anthropol* 2, 65 (1987). [L11] E. Cohen, *Curr. Anthropol.* 53, 588 (2012).

[L12] T. A. Hersh, S. Gero, L. Rendell, and H. Whitehead, *Methods Ecol. Evol.* 12, 1668 (2021), ISSN 2041-210X, 2041-210X.

[L13] S. Gero, A. B ttcher, H. Whitehead, and P. T. Madsen, *R. Soc. Open Sci.* 3, 160061 (2016).

[L14] C. Heyes, *Psychological Bulletin* 137, 463 (2011).

[L15] C. Heyes, *Journal of Comparative Psychology* 126, 193 (2012).

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